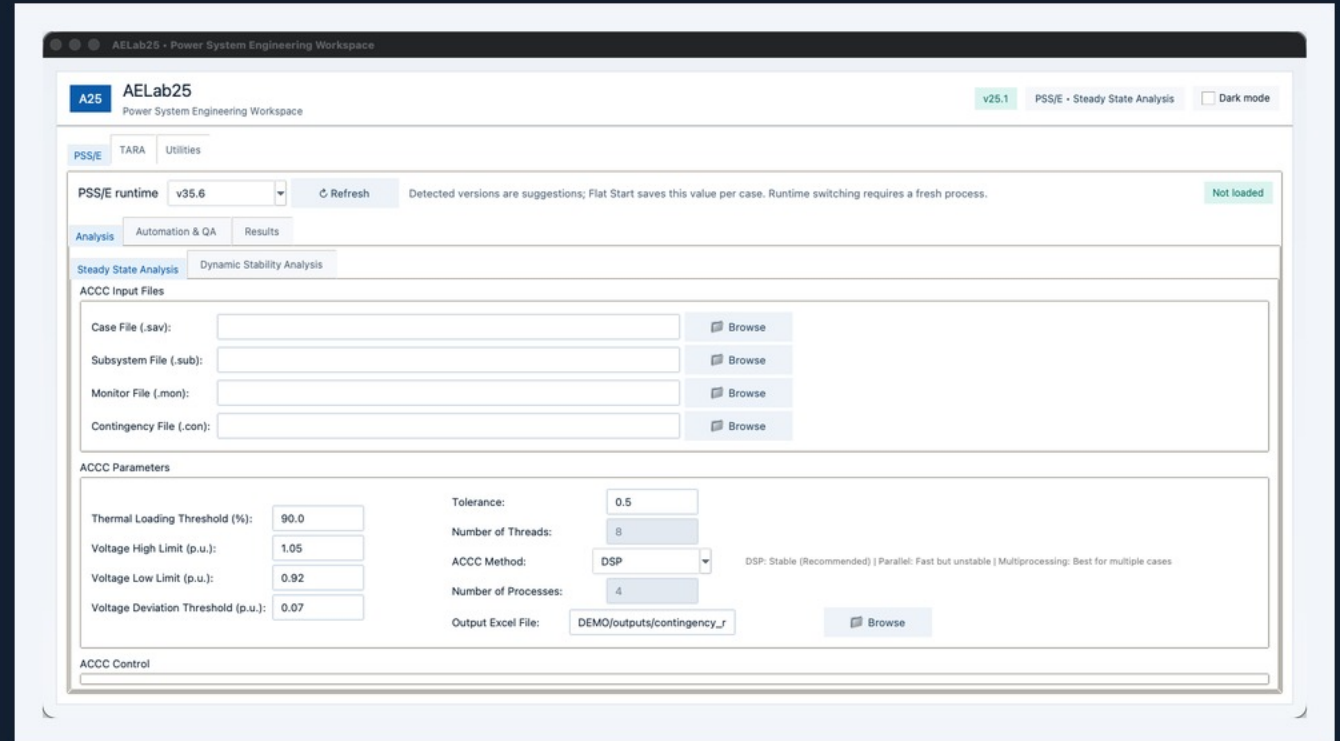


# AELab25 v25.1 Modern UI Overview

## Current workspace and nested-tab documentation

A screen-by-screen guide to the current AELab navigation, configuration, execution, and results workspaces.

Documentation screenshots use blank or synthetic example inputs unless otherwise stated. No licensed PSS/E or TARA solver run, convergence, study result, or engineering validation is represented.



# Workspace map

PSS/E › TARA › Utilities

The catalog follows the visible application hierarchy. Each remaining slide documents one captured workspace or nested tab.

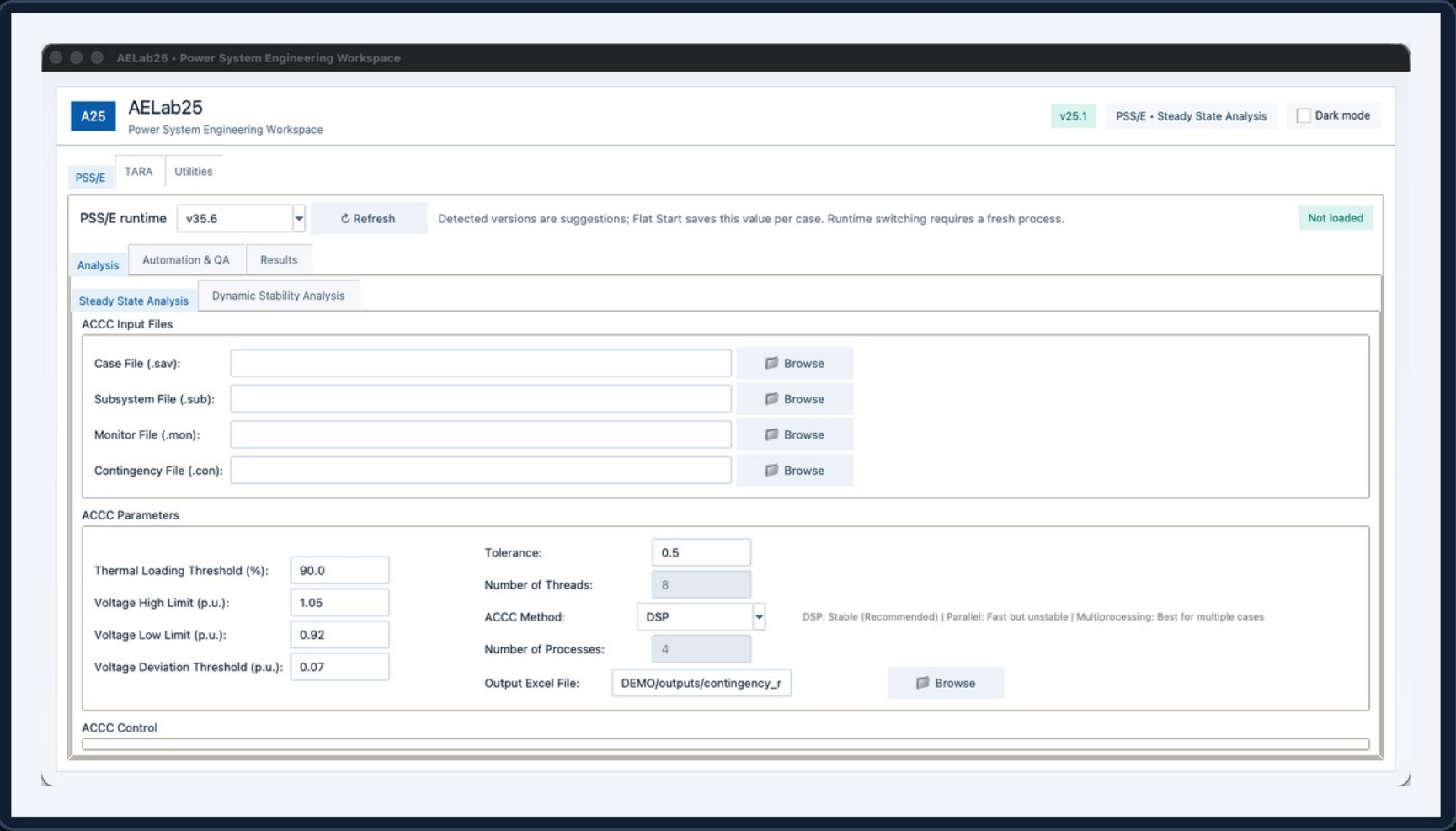
PSS/E — 19 screen(s)

TARA — 14 screen(s)

Utilities — 12 screen(s)

# Steady State Analysis

PSS/E › Analysis › Steady State Analysis



## WHAT THIS TAB DOES

Configures cases, contingency inputs, limits, solver mode, progress, and report export for ACCC studies.

REAL UI CAPTURE

# Flat Start & Validation

PSS/E › Analysis › Dynamic Stability Analysis › Flat Start & Validation

AELab25 • Power System Engineering Workspace

A25

AELab25

Power System Engineering Workspace

v25.1PSS/E • Dynamic Stability AnalysisDark mode

PSS/E

TARA

Utilities

PSS/E runtime

v35.6

Refresh

Detected versions are suggestions; Flat Start saves this value per case. Runtime switching requires a fresh process.

Not loaded

Analysis

Automation & QA

Results

Steady State Analysis

Dynamic Stability Analysis

Flat Start & Validation

Fault Simulation

Dynamic Simulator

Flat-Start Case Preparation

Prepare converted cases and snapshots, verify a zero-disturbance flat run, measure deviations, and create an interactive AutoPlot dashboard.

Study Configuration

Cases root

DEMO

Browse

Case config

Discover

Python executable

python

Browse

New Config

Save

Reload

Case

Scenario

Flat Start

Choose a case configuration.

Resolved Plan

Read-only preflight snapshot; commands execute without a shell.

Clear

Copy

Live Run Log

Driver output and explicit AELAB\_STAGE progress markers.

Clear

Copy

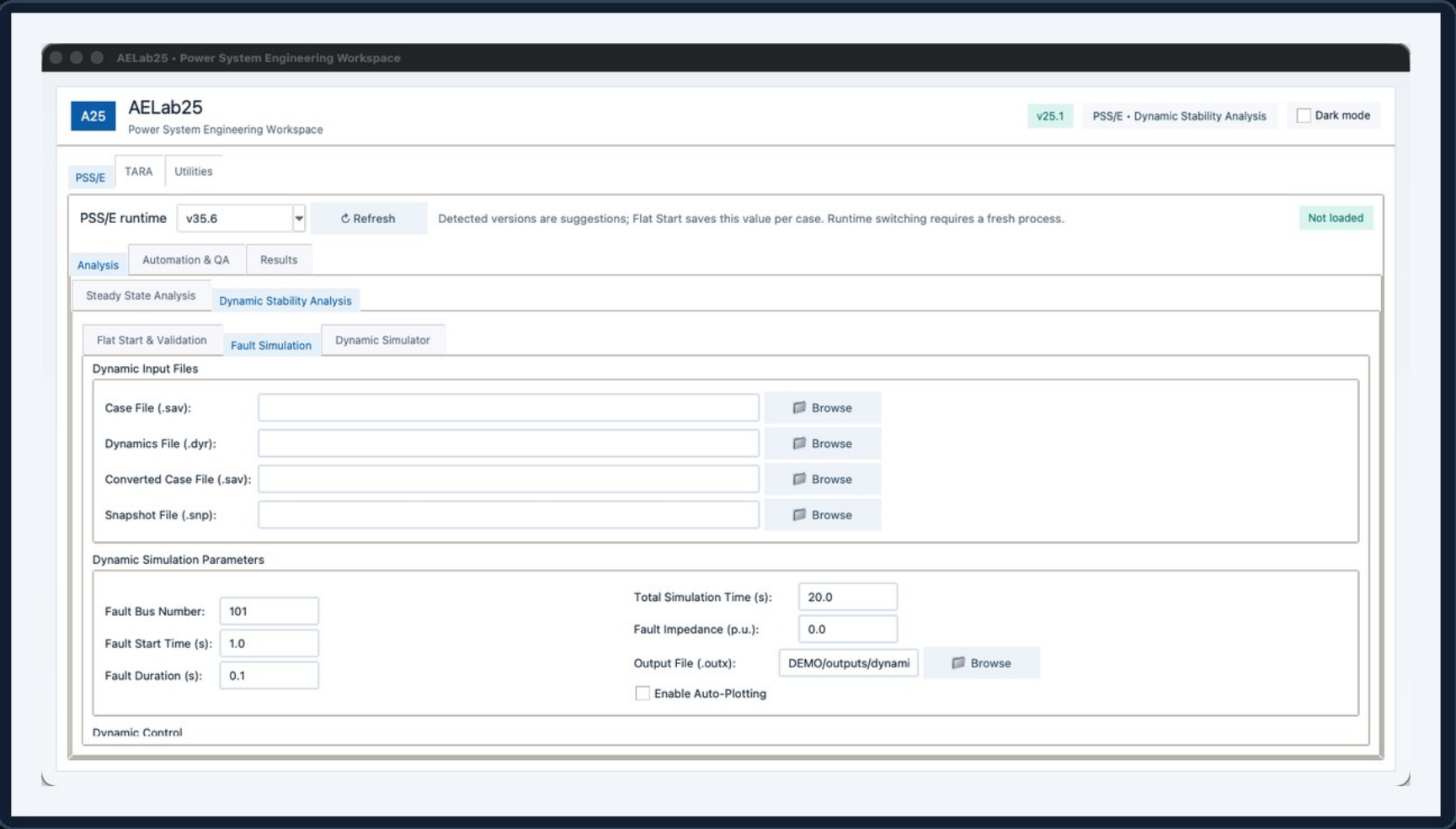
## WHAT THIS TAB DOES

Prepares converted cases and snapshots, verifies a zero-disturbance run, checks deviations, and can build an AutoPlot dashboard.

REAL UI CAPTURE

# Fault Simulation

PSS/E › Analysis › Dynamic Stability Analysis › Fault Simulation



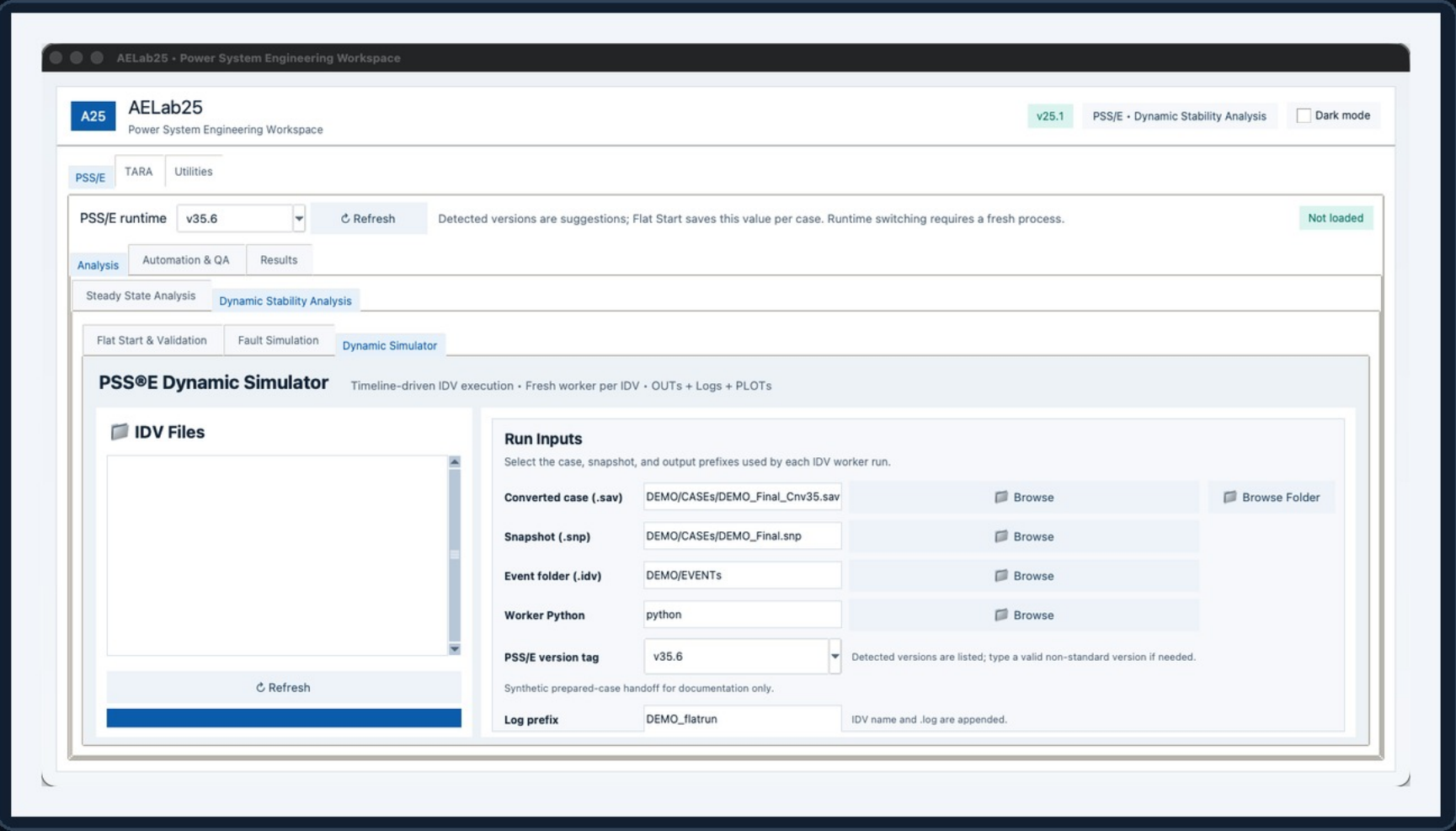
## WHAT THIS TAB DOES

Collects prepared dynamic inputs, fault timing and impedance, simulation duration, output path, progress, and optional plotting.

REAL UI CAPTURE

# Dynamic Simulator

PSS/E › Analysis › Dynamic Stability Analysis › Dynamic Simulator



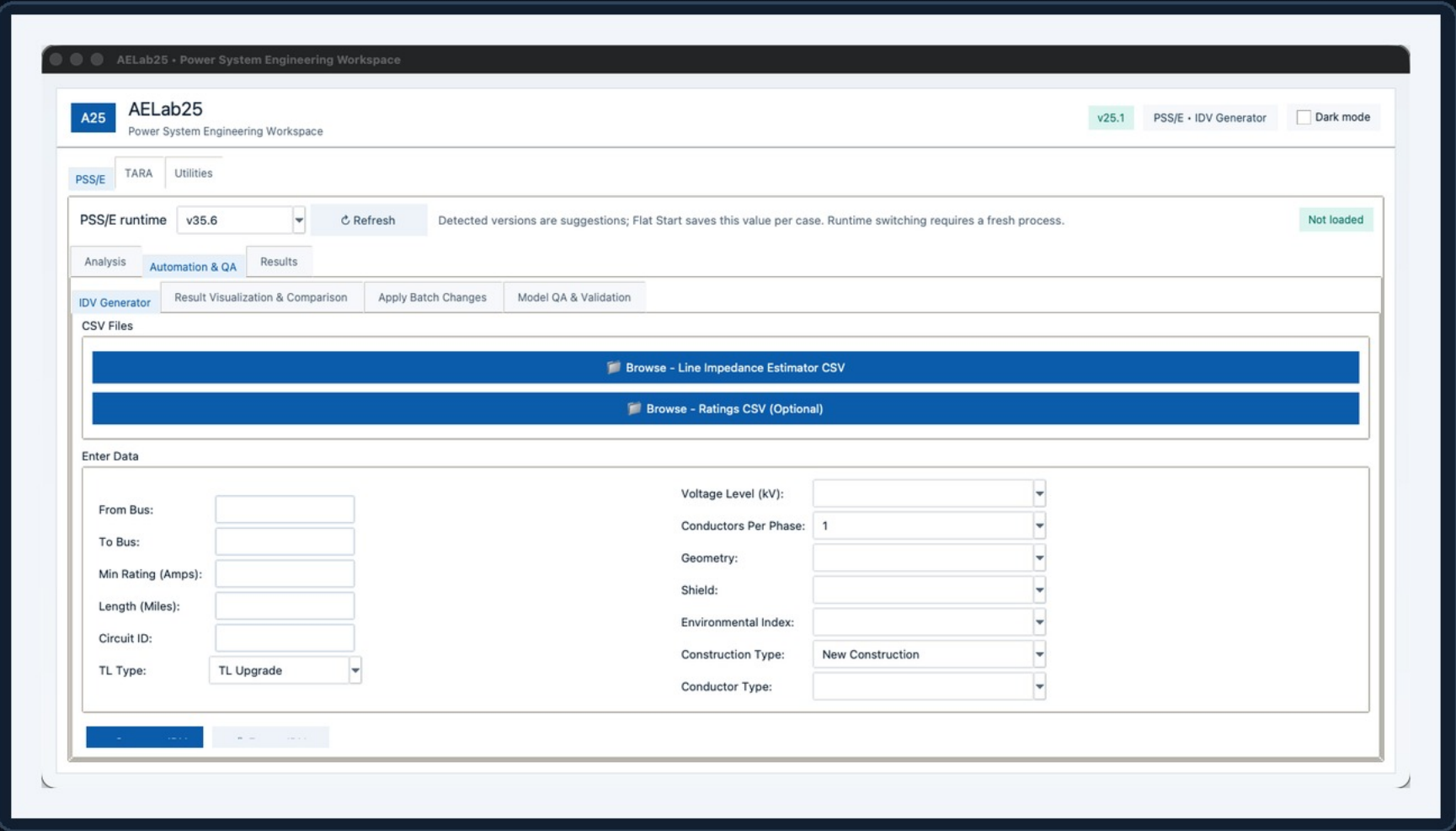
## WHAT THIS TAB DOES

Provides the GUI workflow for prepared-case IDV event batches, version-aware execution settings, status, and output review.

REAL UI CAPTURE

# IDV Generator

PSS/E › Automation & QA › IDV Generator



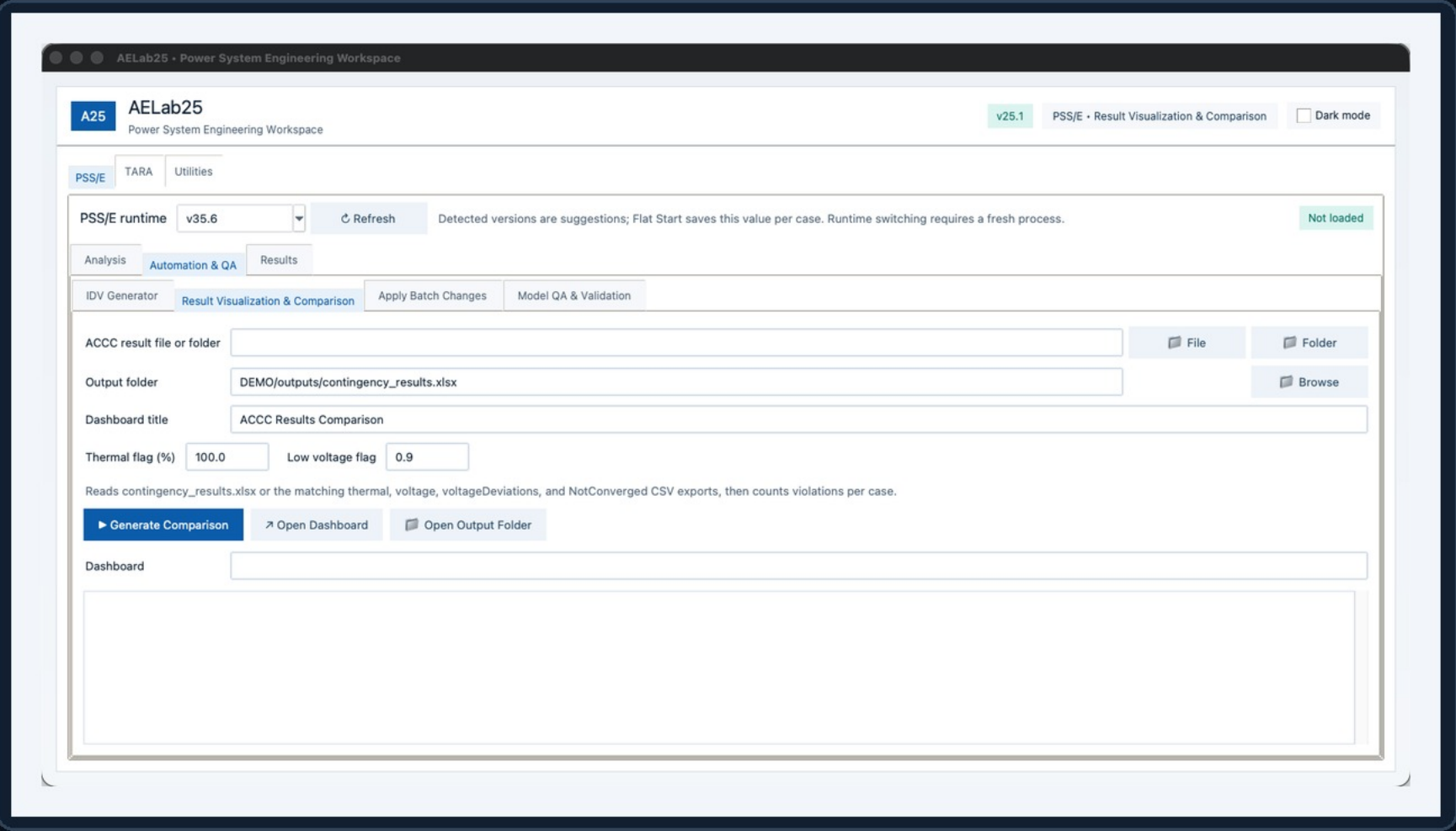
## WHAT THIS TAB DOES

Builds transmission-line change commands from impedance, rating, geometry, construction, and circuit inputs.

REAL UI CAPTURE

# Result Visualization & Comparison

PSS/E › Automation & QA › Result Visualization & Comparison



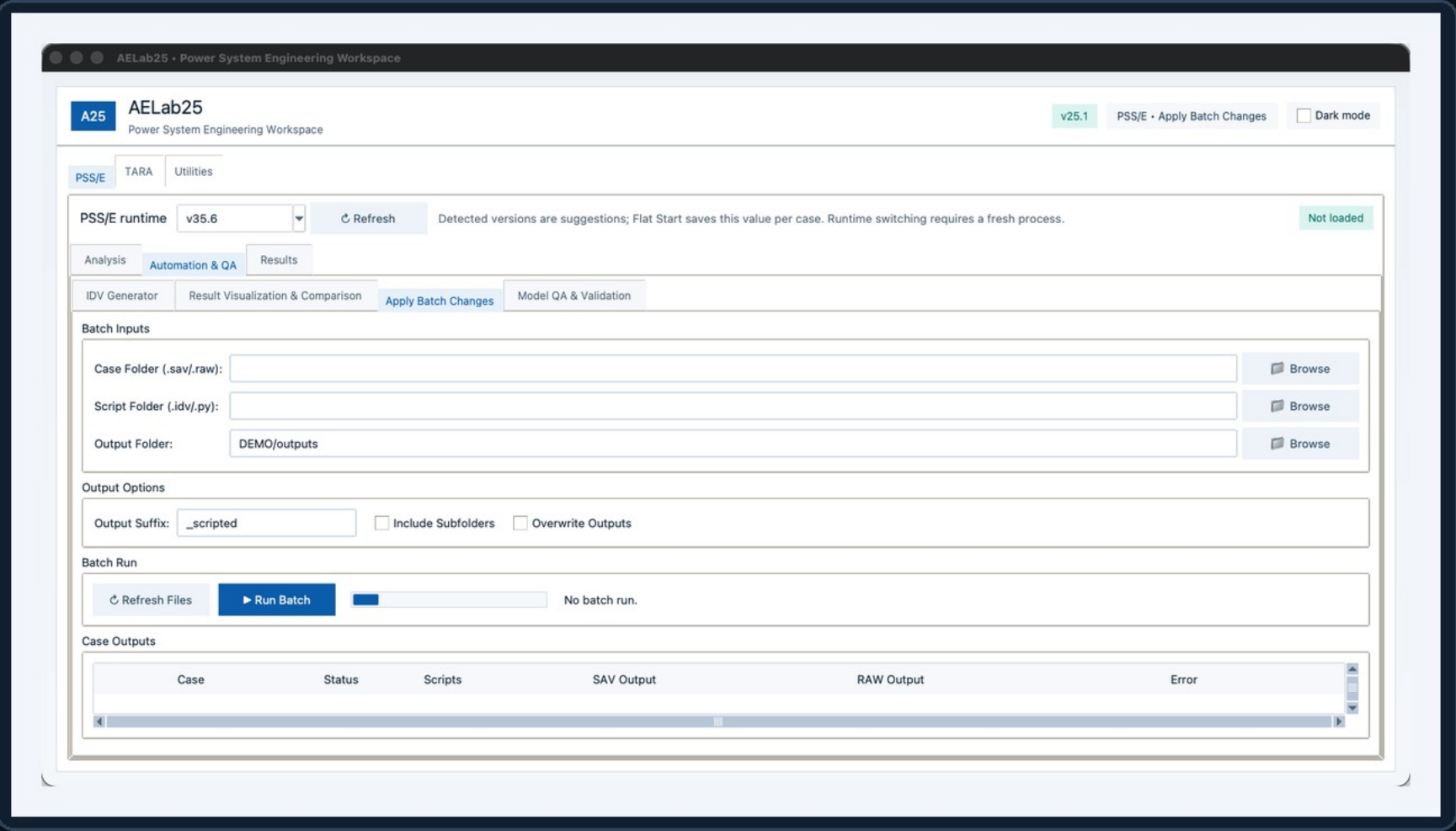
## WHAT THIS TAB DOES

Combines ACCC result files into a filterable comparison dashboard with configurable thermal and voltage flags.

REAL UI CAPTURE

# Apply Batch Changes

PSS/E > Automation & QA > Apply Batch Changes



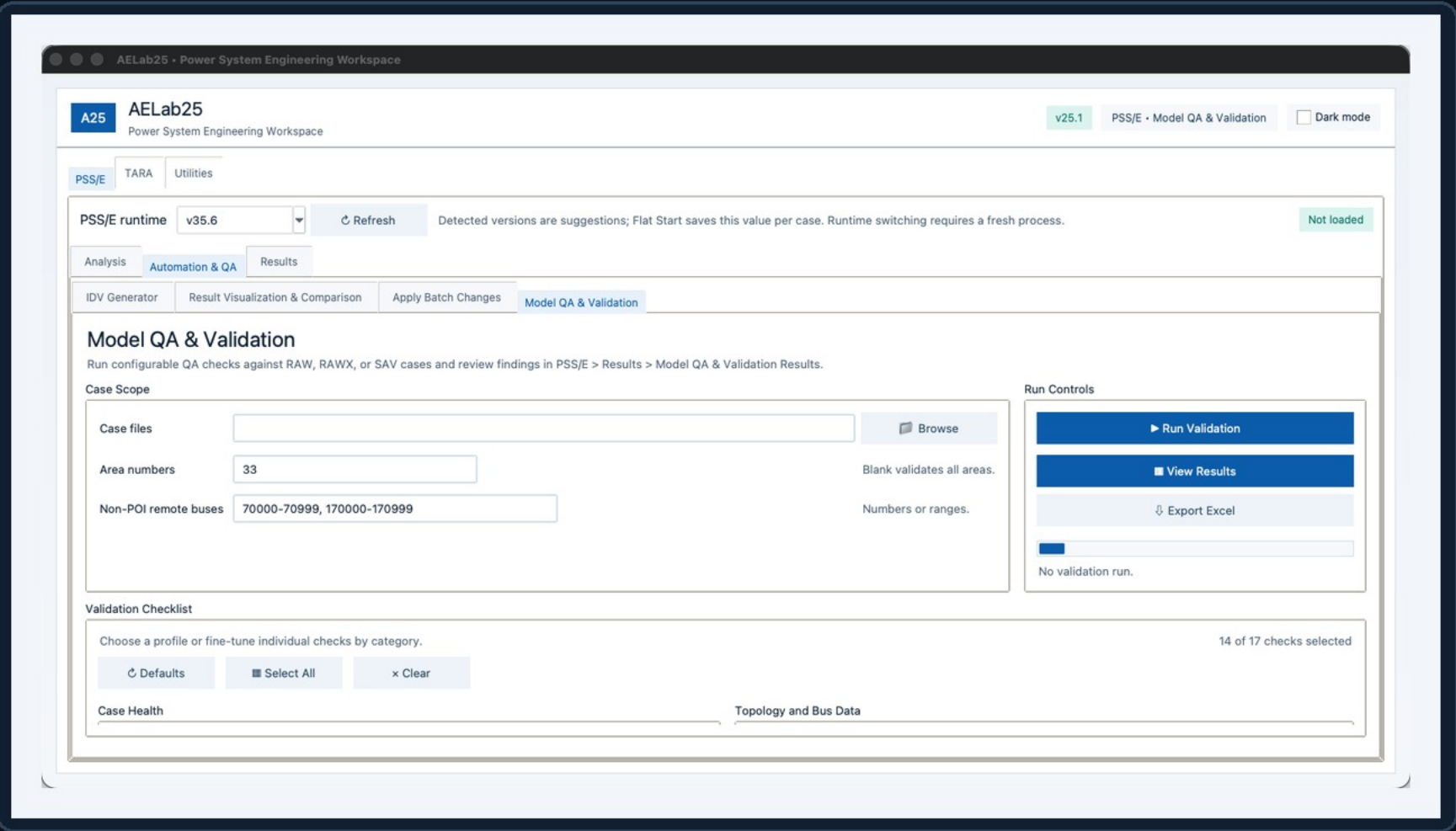
## WHAT THIS TAB DOES

Applies selected case scripts to a folder of source cases and reports derived outputs without overwriting originals by default.

REAL UI CAPTURE

# Model QA & Validation

PSS/E › Automation & QA › Model QA & Validation



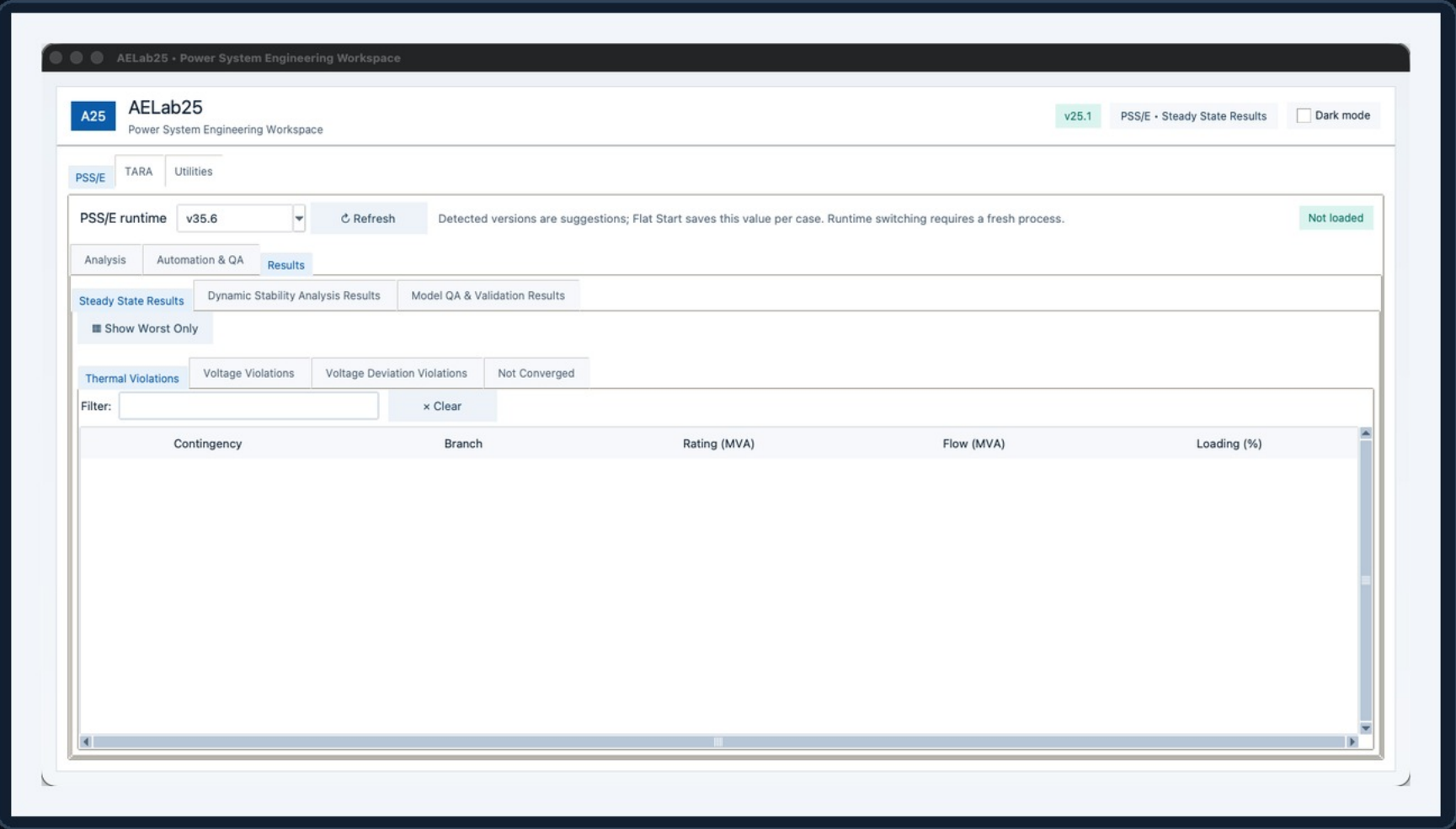
## WHAT THIS TAB DOES

Selects case scope and deterministic model checks, then records findings for engineering review.

REAL UI CAPTURE

# Thermal Violations

PSS/E › Results › Steady State Results › Thermal Violations



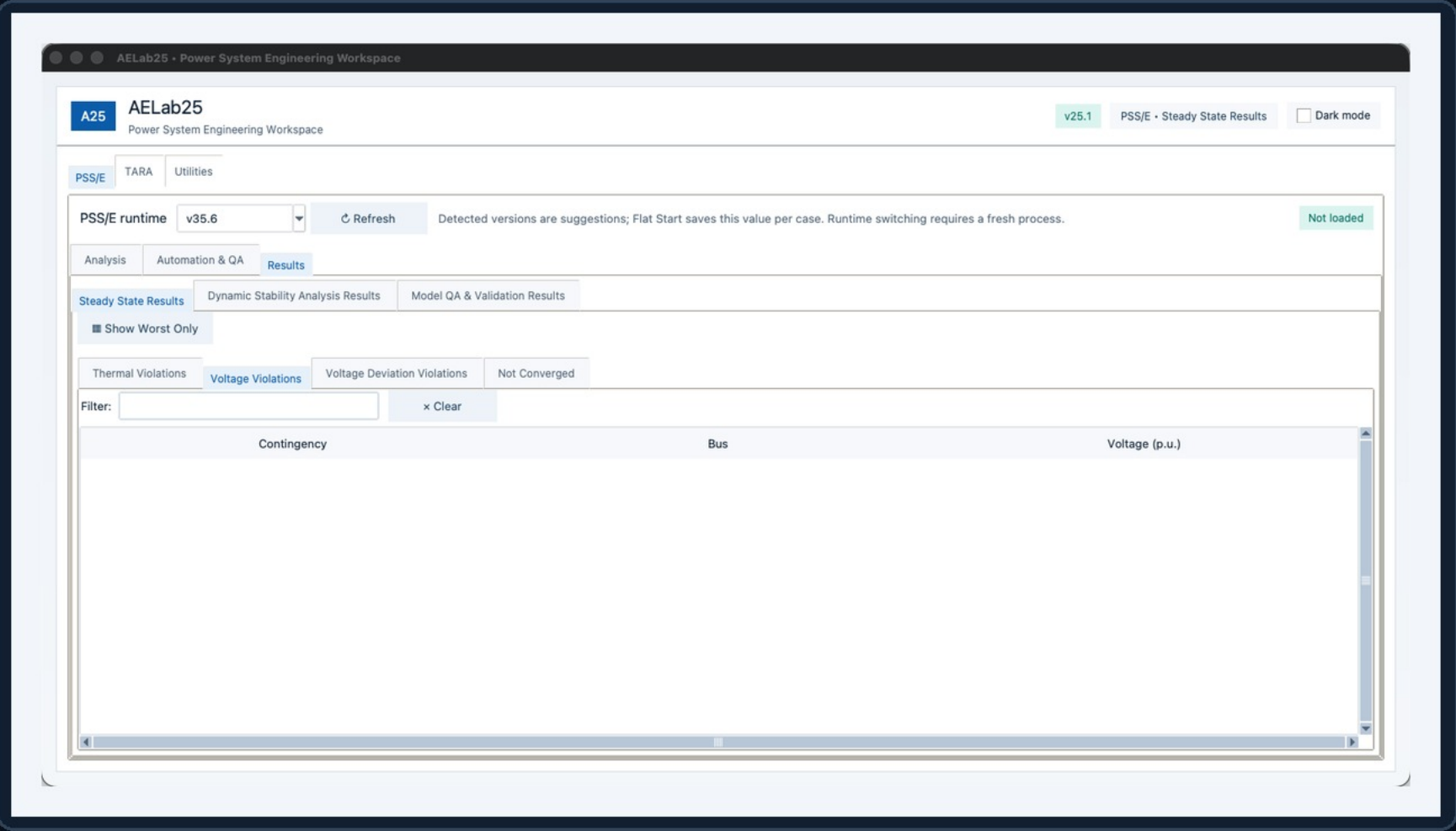
## WHAT THIS TAB DOES

Reviews contingency, branch, rating, flow, and loading columns for reported thermal violations.

REAL UI CAPTURE

# Voltage Violations

PSS/E › Results › Steady State Results › Voltage Violations



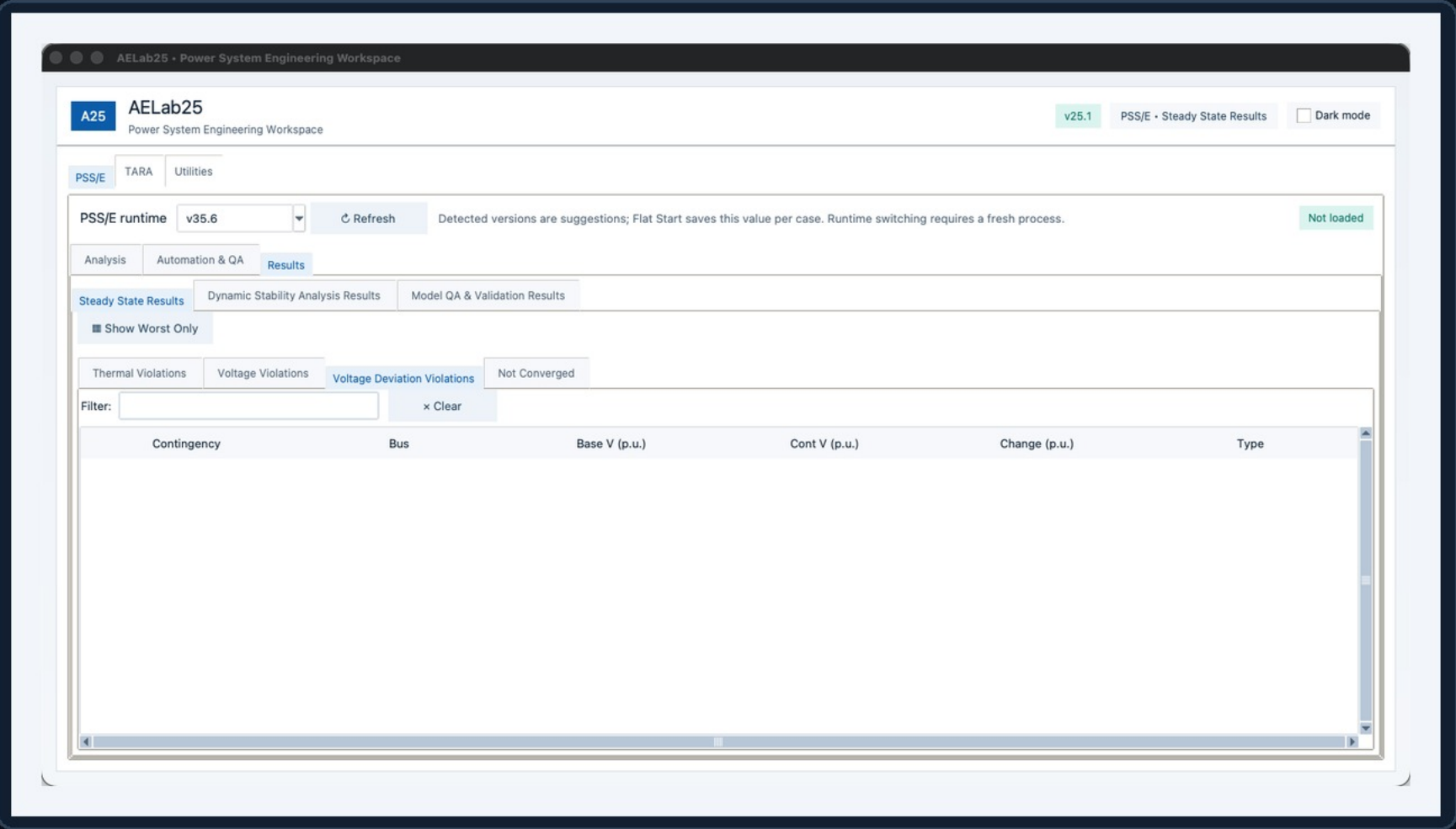
## WHAT THIS TAB DOES

Reviews reported contingency bus voltages in per unit.

REAL UI CAPTURE

# Voltage Deviation Violations

PSS/E › Results › Steady State Results › Voltage Deviation Violations



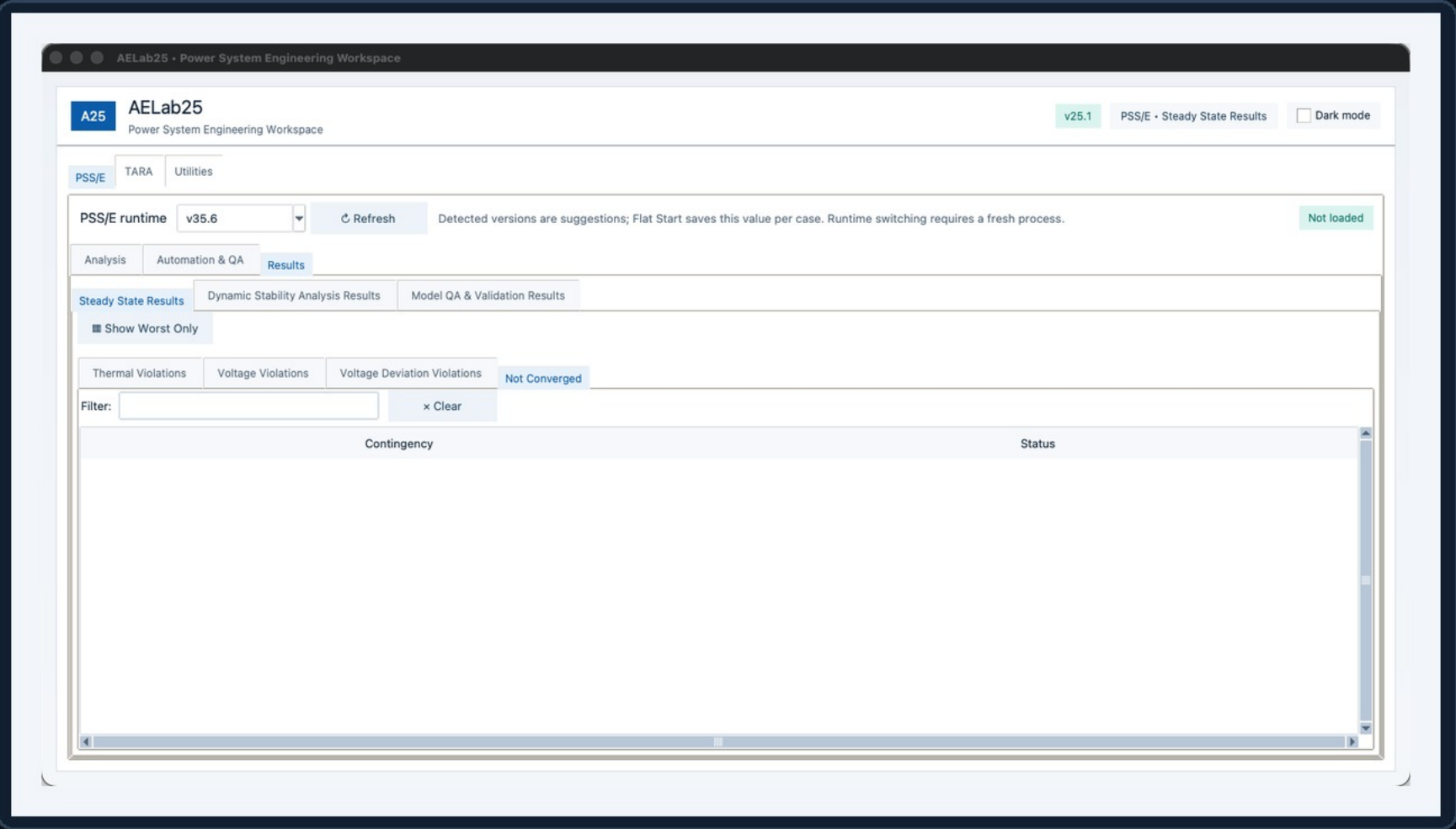
## WHAT THIS TAB DOES

Compares base and contingency voltage and identifies the reported rise or drop magnitude.

REAL UI CAPTURE

# Not Converged

PSS/E › Results › Steady State Results › Not Converged



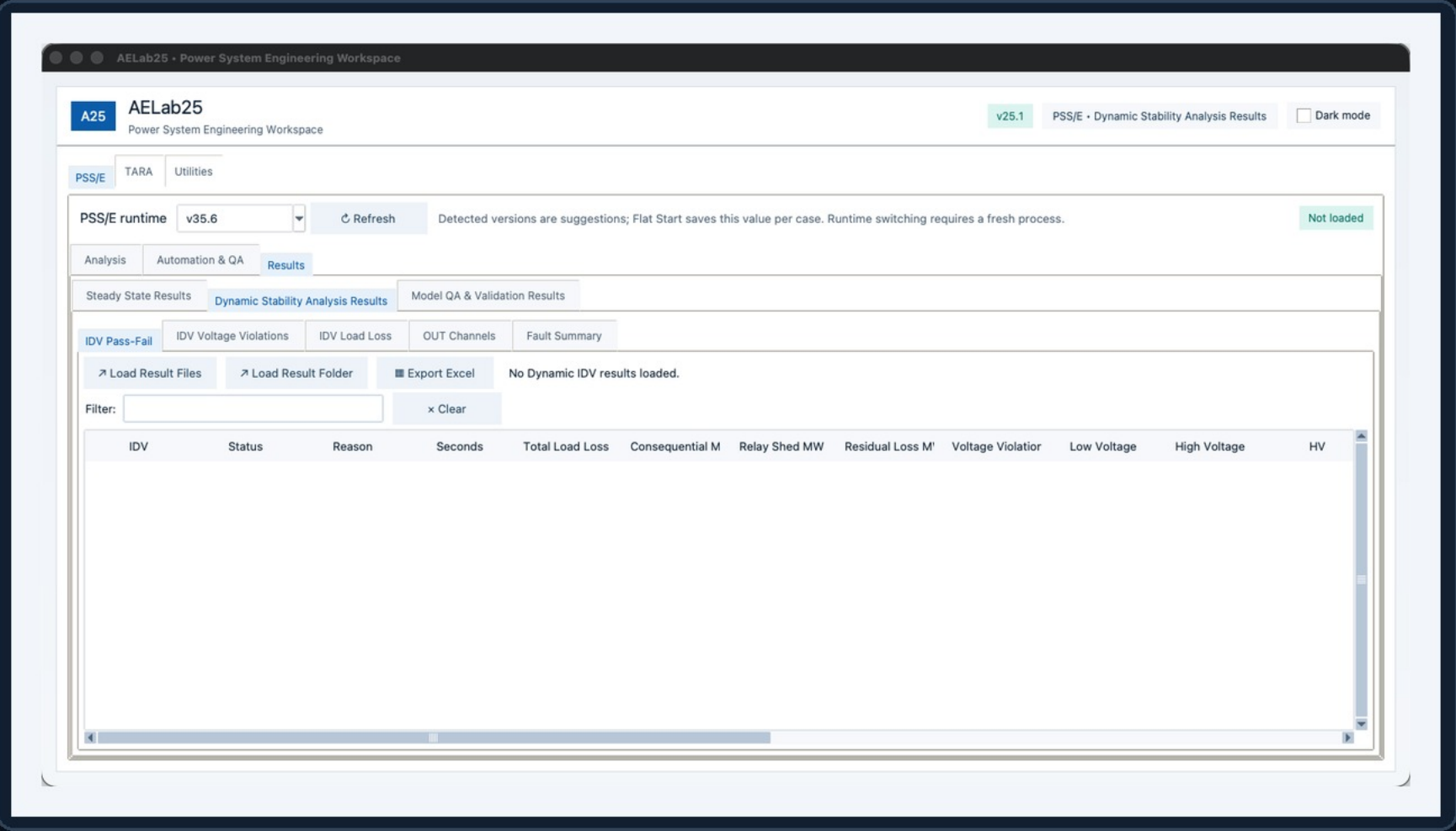
## WHAT THIS TAB DOES

Keeps nonconverged contingency outcomes explicit instead of treating them as passing cases.

REAL UI CAPTURE

# IDV Pass-Fail

PSS/E › Results › Dynamic Stability Analysis Results › IDV Pass-Fail



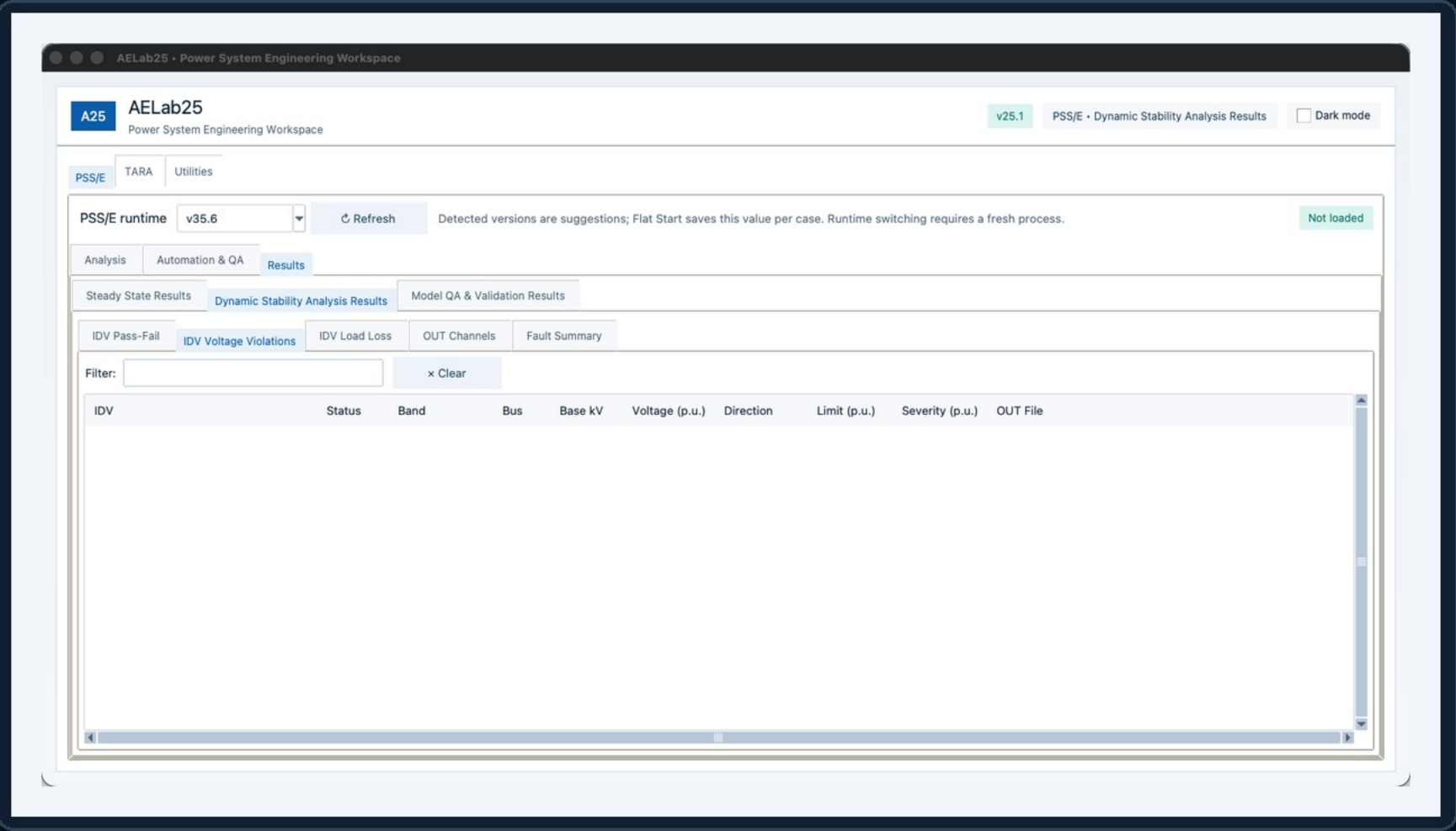
## WHAT THIS TAB DOES

Summarizes each IDV result, reason, elapsed time, load loss, voltage counts, and source artifacts.

REAL UI CAPTURE

# IDV Voltage Violations

PSS/E › Results › Dynamic Stability Analysis Results › IDV Voltage Violations



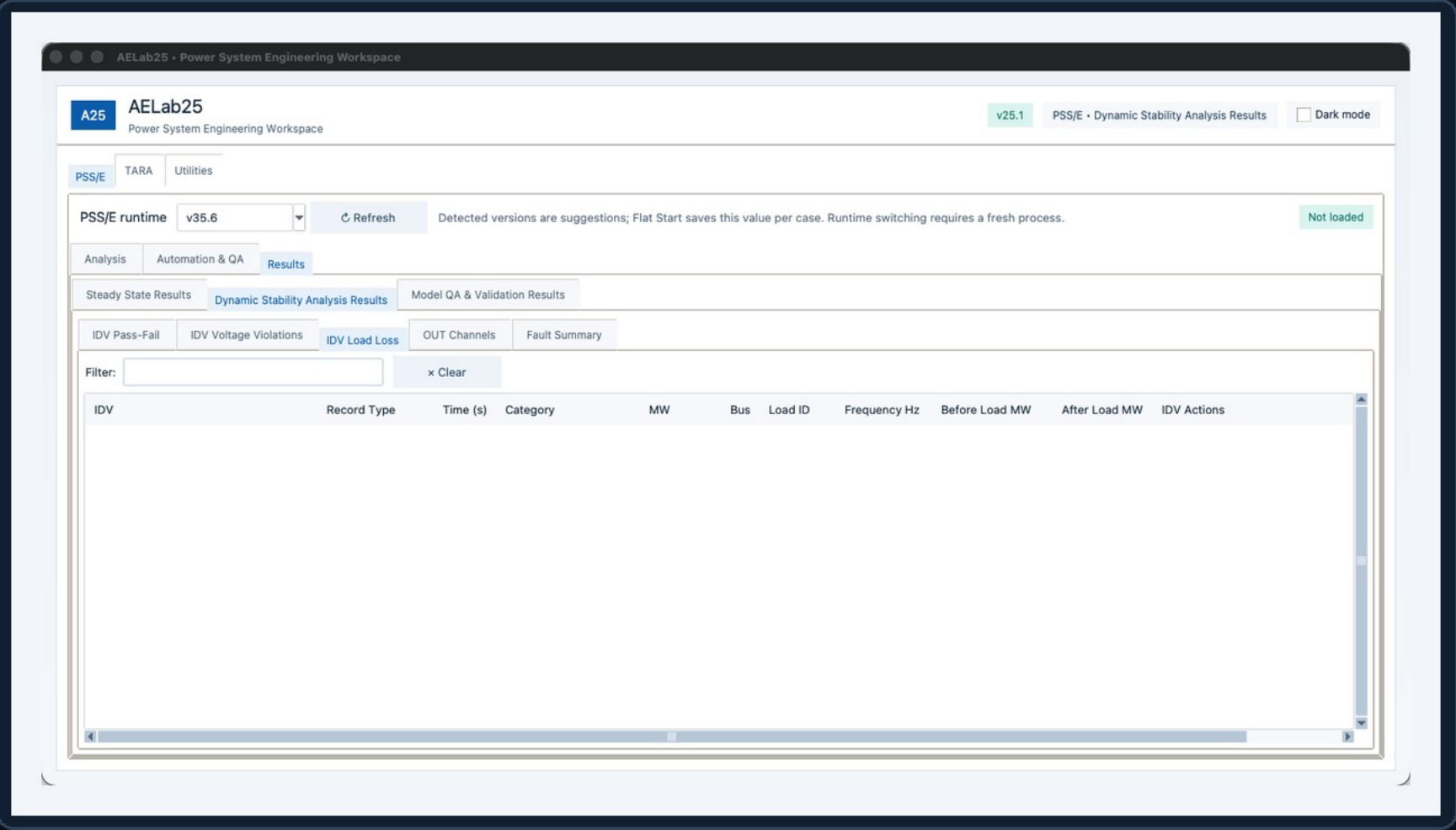
## WHAT THIS TAB DOES

Reviews low and high dynamic voltage observations by band, bus, limit, and severity.

REAL UI CAPTURE

# IDV Load Loss

PSS/E › Results › Dynamic Stability Analysis Results › IDV Load Loss



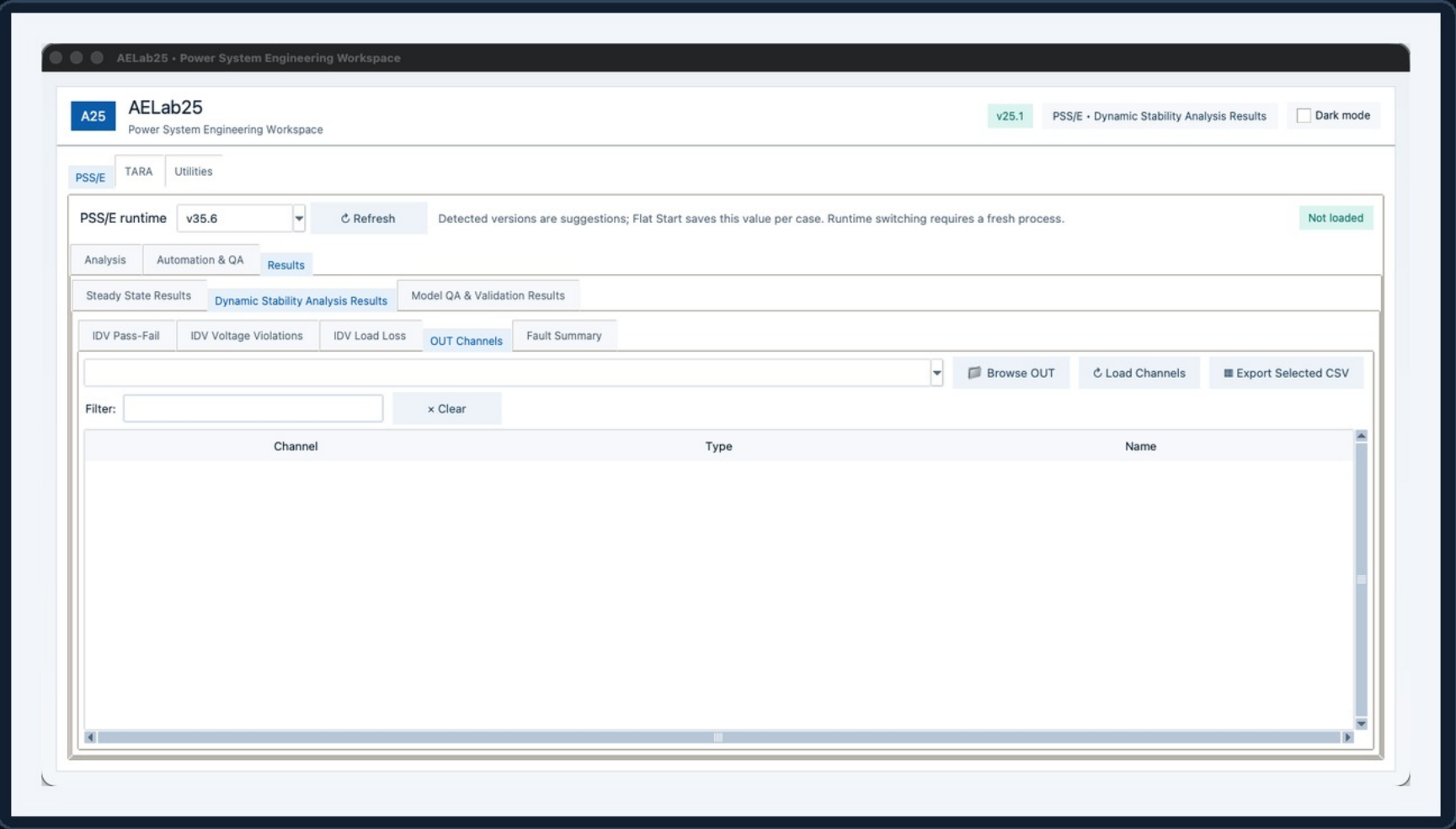
## WHAT THIS TAB DOES

Reviews load-loss records with time, category, MW, bus, frequency, before/after load, and IDV actions.

REAL UI CAPTURE

# OUT Channels

PSS/E › Results › Dynamic Stability Analysis Results › OUT Channels



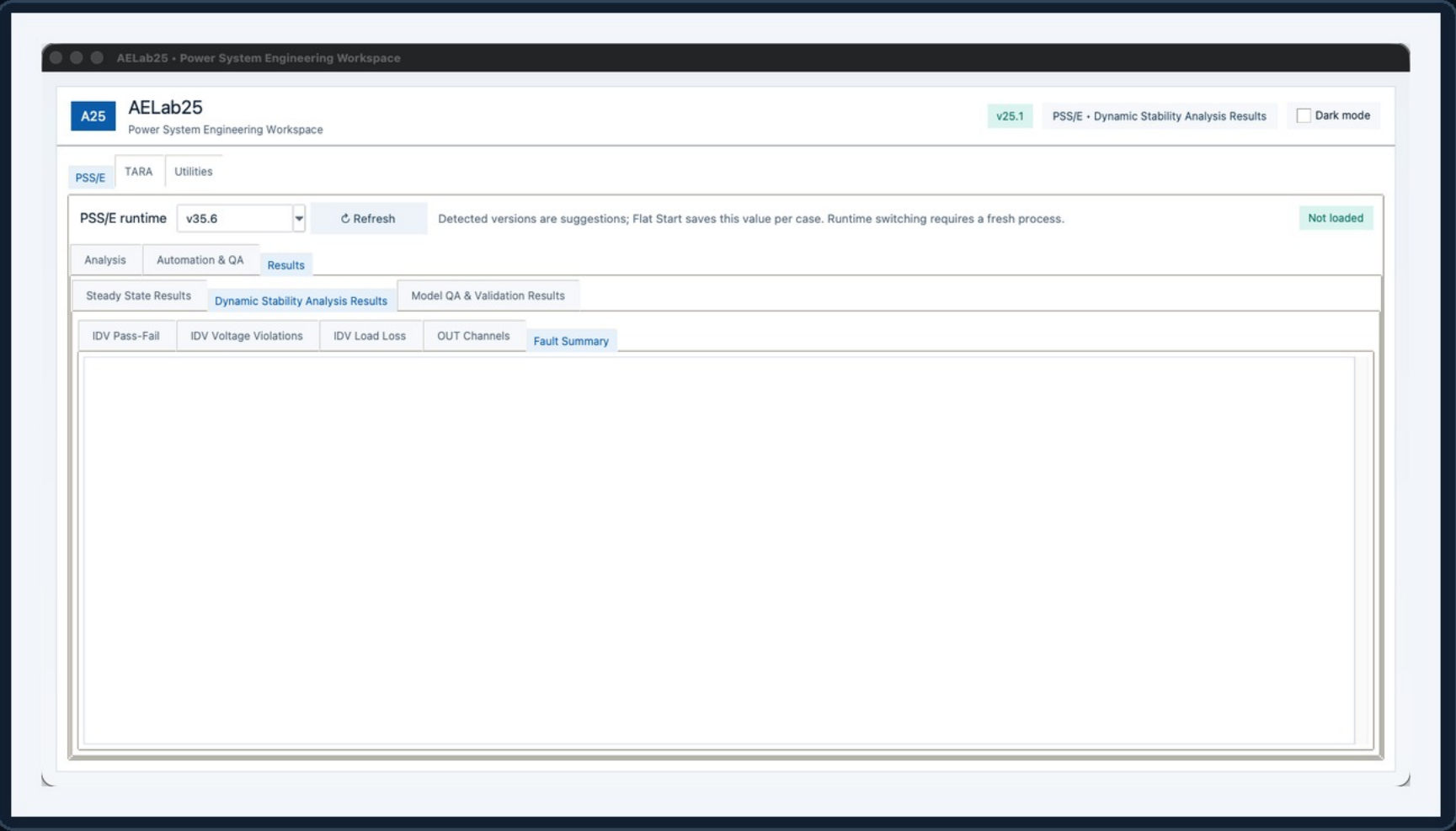
## WHAT THIS TAB DOES

Loads an OUT file channel index and exports selected channel data to CSV.

REAL UI CAPTURE

# Fault Summary

PSS/E › Results › Dynamic Stability Analysis Results › Fault Summary



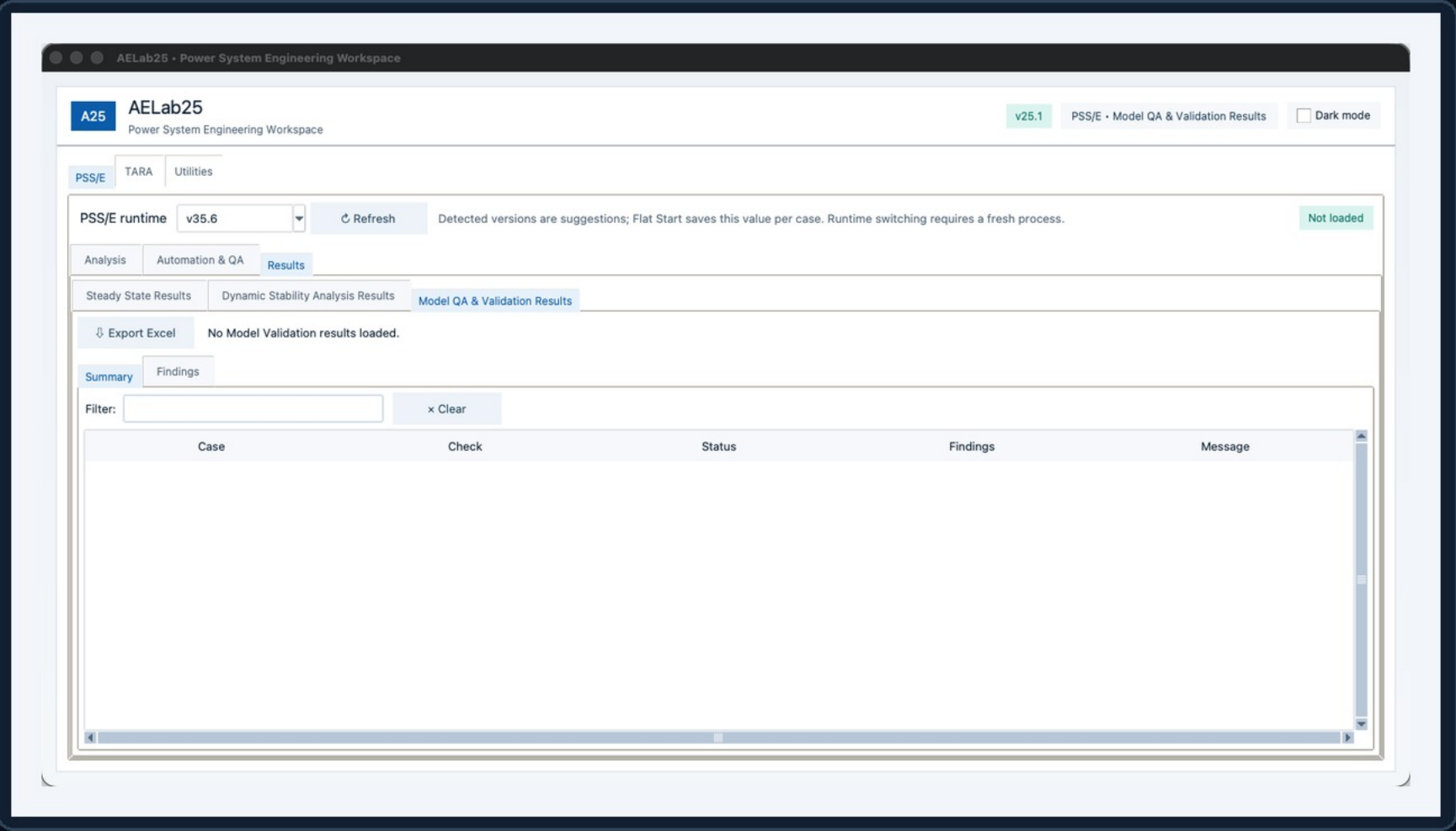
## WHAT THIS TAB DOES

Displays the text summary produced by the fault-simulation workflow.

REAL UI CAPTURE

# Model QA Summary

PSS/E › Results › Model QA & Validation Results › Summary



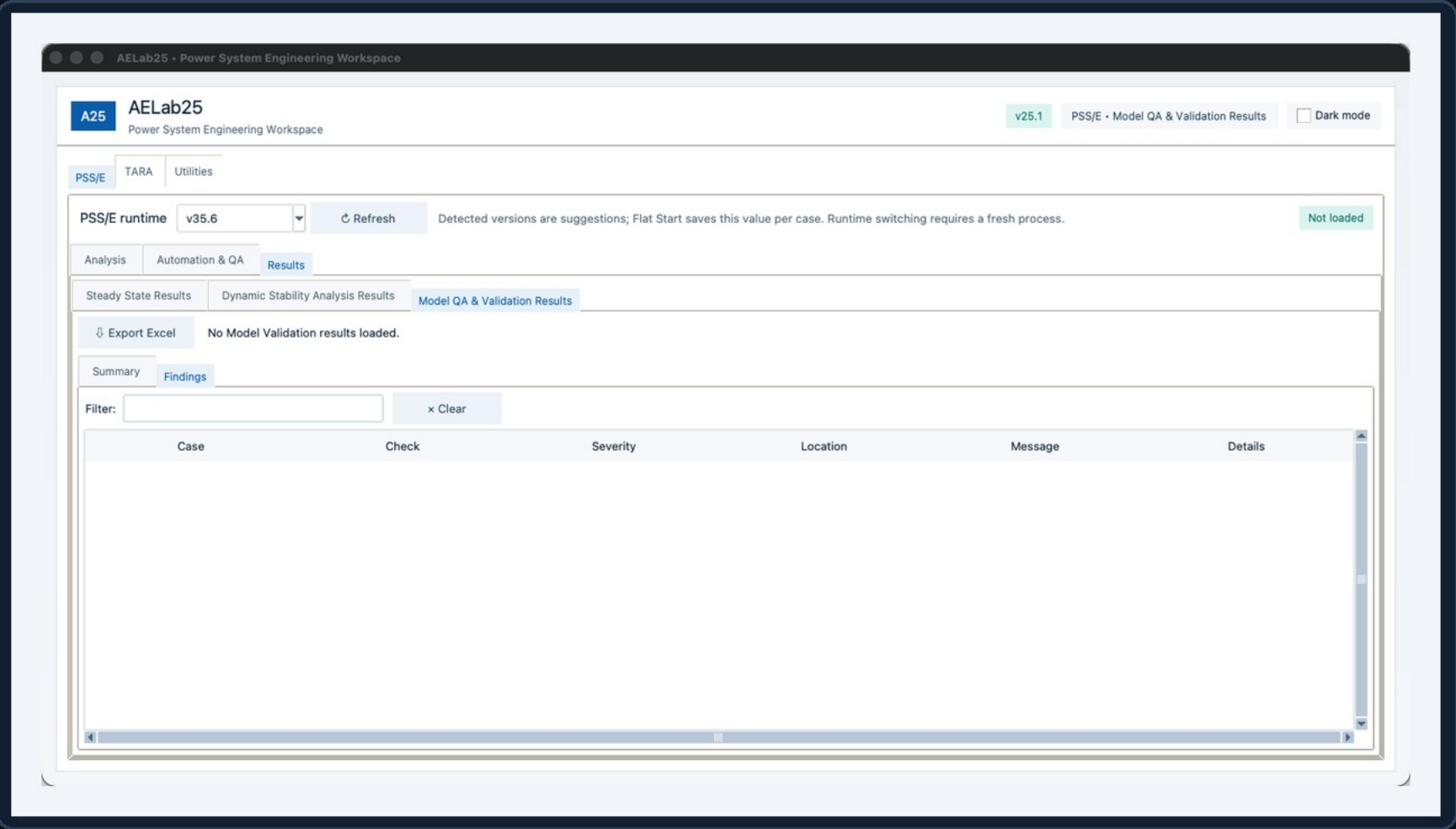
## WHAT THIS TAB DOES

Summarizes validation status and finding counts by case and check.

REAL UI CAPTURE

# Model QA Findings

PSS/E › Results › Model QA & Validation Results › Findings



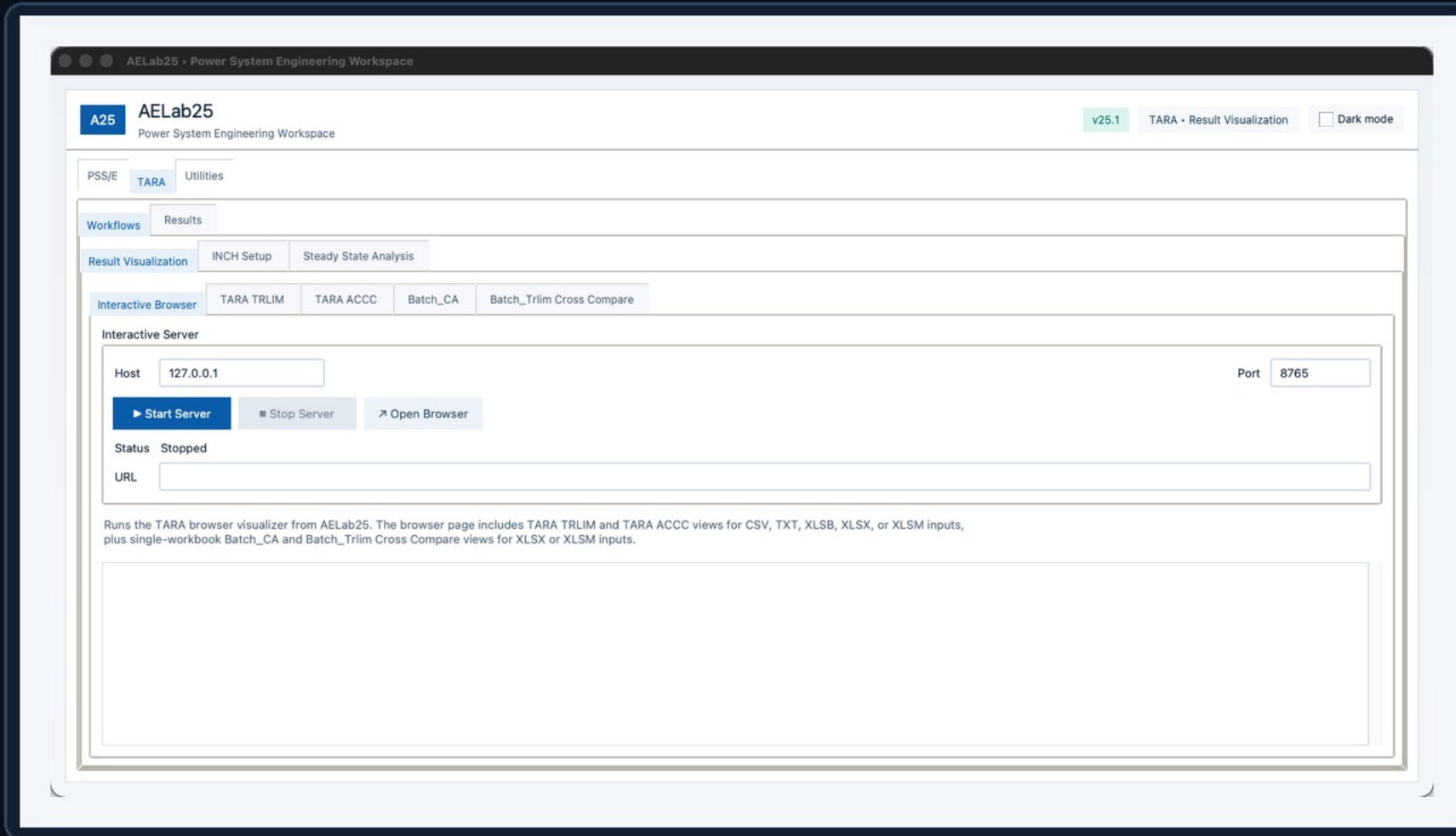
## WHAT THIS TAB DOES

Lists individual findings with severity, location, message, and supporting details.

REAL UI CAPTURE

# Interactive Browser

TARA › Workflows › Result Visualization › Interactive Browser



## WHAT THIS TAB DOES

Starts and links to the local interactive TARA result browser without embedding source study data in the GUI.

## REAL UI CAPTURE

# TARA TRLIM

TARA › Workflows › Result Visualization › TARA TRLIM

The screenshot shows the TARA TRLIM configuration interface within the AELab25 Power System Engineering Workspace. The interface includes a top navigation bar with the workspace name, version (v25.1), and a dark mode toggle. Below this, there are tabs for PSS/E, TARA, and Utilities. The TARA tab is active, showing sub-tabs for Workflows and Results. Under Results, there are tabs for Result Visualization, INCH Setup, and Steady State Analysis. The Result Visualization tab is active, showing a sub-tab for Interactive Browser. Under Interactive Browser, there are tabs for TARA TRLIM, TARA ACCC, Batch\_CA, and Batch\_Trlim Cross Compare. The TARA TRLIM tab is active, showing a form with fields for TARA result file, Excel sheet, Output folder, and Dashboard title. There are also input fields for Top items (15) and Table row limit (2000). At the bottom, there are buttons for Generate Dashboard, Open Dashboard, and Open Output Folder.

AE Lab25 • Power System Engineering Workspace

A25 AE Lab25 Power System Engineering Workspace v25.1 TARA • Result Visualization Dark mode

PSS/E TARA Utilities

Workflows Results

Result Visualization INCH Setup Steady State Analysis

Interactive Browser TARA TRLIM TARA ACCC Batch\_CA Batch\_Trlim Cross Compare

TARA result file  Browse

Excel sheet  Optional for XLSB/XLSX

Output folder DEMO/outputs  Browse

Dashboard title

Top items 15 Table row limit 2000

► Generate Dashboard ↗ Open Dashboard 📁 Open Output Folder

Dashboard

## WHAT THIS TAB DOES

Configures a searchable HTML dashboard for a TARA transfer-limit result file.

REAL UI CAPTURE

# TARA ACCC

TARA › Workflows › Result Visualization › TARA ACCC

AE Lab25 • Power System Engineering Workspace

A25

AE Lab25

Power System Engineering Workspace

v25.1TARA • Result VisualizationDark mode

PSS/E

TARA

Utilities

Workflows

Results

Result Visualization

INCH Setup

Steady State Analysis

Interactive Browser

TARA TRLIM

TARA ACCC

Batch\_CA

Batch\_Trlim Cross Compare

TARAVIEWER ACCC WORKBOOK

Browse

Output folder

DEMO/outputs/contingency\_results.xlsx

Browse

DASHBOARD TITLE

TARA ACCC Results Comparison

Thermal flag (%)

100.0

Low voltage flag

0.9

Reads TARAVIEWER ACCC XLSX sheets BranchViol, VoltViol, and CaseSum, then compares branch and voltage violation counts per LF case.

Generate Comparison

Open Dashboard

Open Output Folder

DASHBOARD

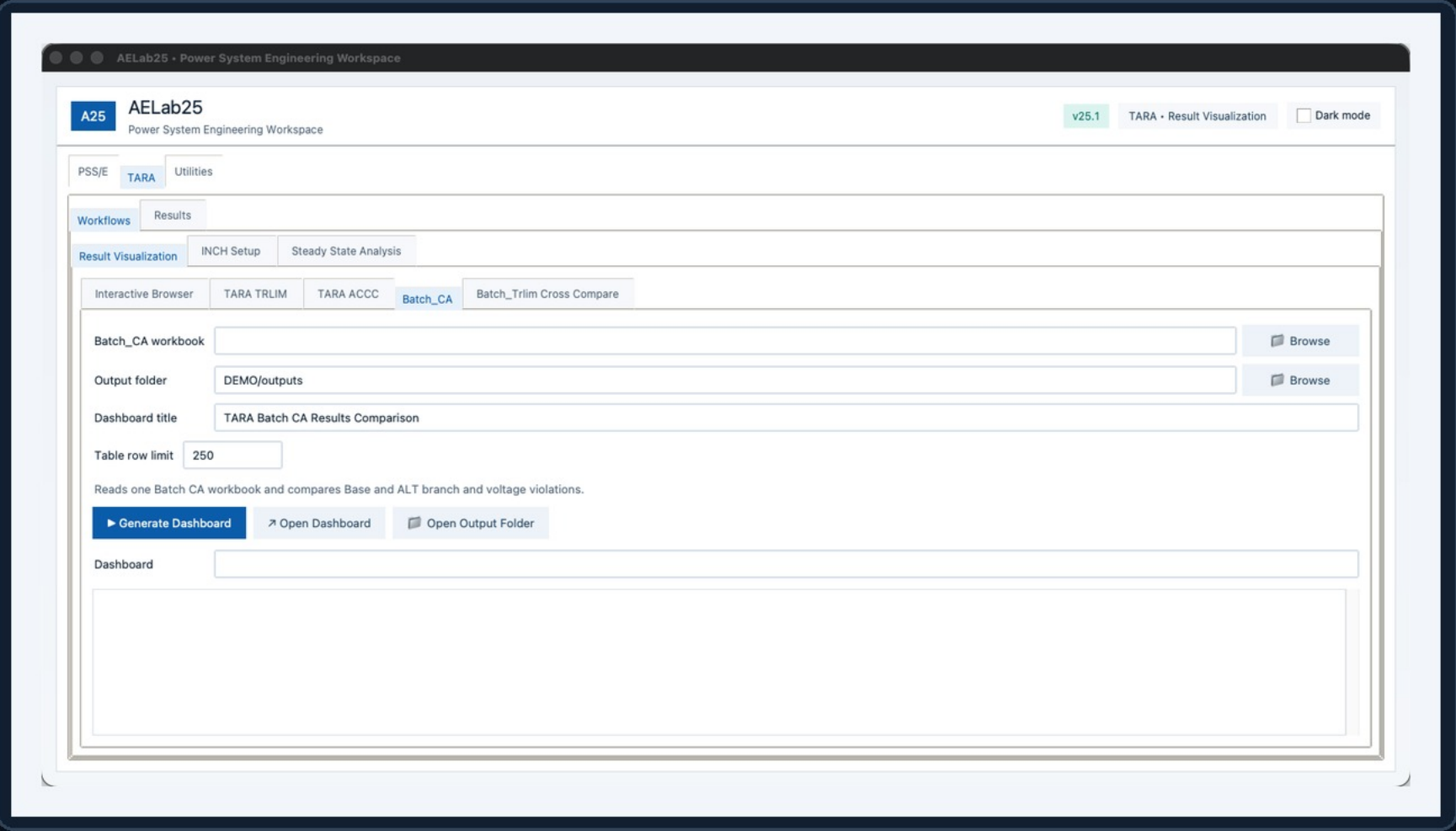
## WHAT THIS TAB DOES

Configures an ACCC result dashboard with explicit thermal and low-voltage review flags.

REAL UI CAPTURE

# Batch\_CA

TARA › Workflows › Result Visualization › Batch\_CA



## WHAT THIS TAB DOES

Builds a combined browser from a folder of TARA contingency-analysis result files.

REAL UI CAPTURE

# Batch\_Trlim Cross Compare

TARA › Workflows › Result Visualization › Batch\_Trlim Cross Compare

The screenshot displays the AELab25 Power System Engineering Workspace interface. The top navigation bar includes the workspace name 'AELab25', version 'v25.1', and the current tab 'TARA • Result Visualization'. A 'Dark mode' toggle is present. The left sidebar shows a hierarchy: 'PSS/E' > 'TARA' > 'Utilities'. The main content area has tabs for 'Workflows' and 'Results', with 'Result Visualization' selected. Under 'Result Visualization', there are sub-tabs: 'INCH Setup', 'Steady State Analysis', and 'Batch\_Trlim Cross Compare'. The 'Batch\_Trlim Cross Compare' tab is active, showing a configuration form with fields for 'Batch\_Trlim Cross Compare workbook', 'Output folder', 'Dashboard title', and 'Table row limit'. Below these fields is a descriptive text and three buttons: 'Generate Dashboard', 'Open Dashboard', and 'Open Output Folder'. The 'Dashboard' section at the bottom is currently empty.

AELab25 Power System Engineering Workspace

v25.1 TARA • Result Visualization Dark mode

PSS/E TARA Utilities

Workflows Results

Result Visualization INCH Setup Steady State Analysis

Interactive Browser TARA TRLIM TARA ACCC Batch\_CA Batch\_Trlim Cross Compare

Batch\_Trlim Cross Compare workbook  Browse

Output folder  Browse

Dashboard title

Table row limit

Reads one Batch TRLIM workbook and cross-compares its Base and ALT branch and voltage pivot results.

► Generate Dashboard ↗ Open Dashboard 📁 Open Output Folder

Dashboard

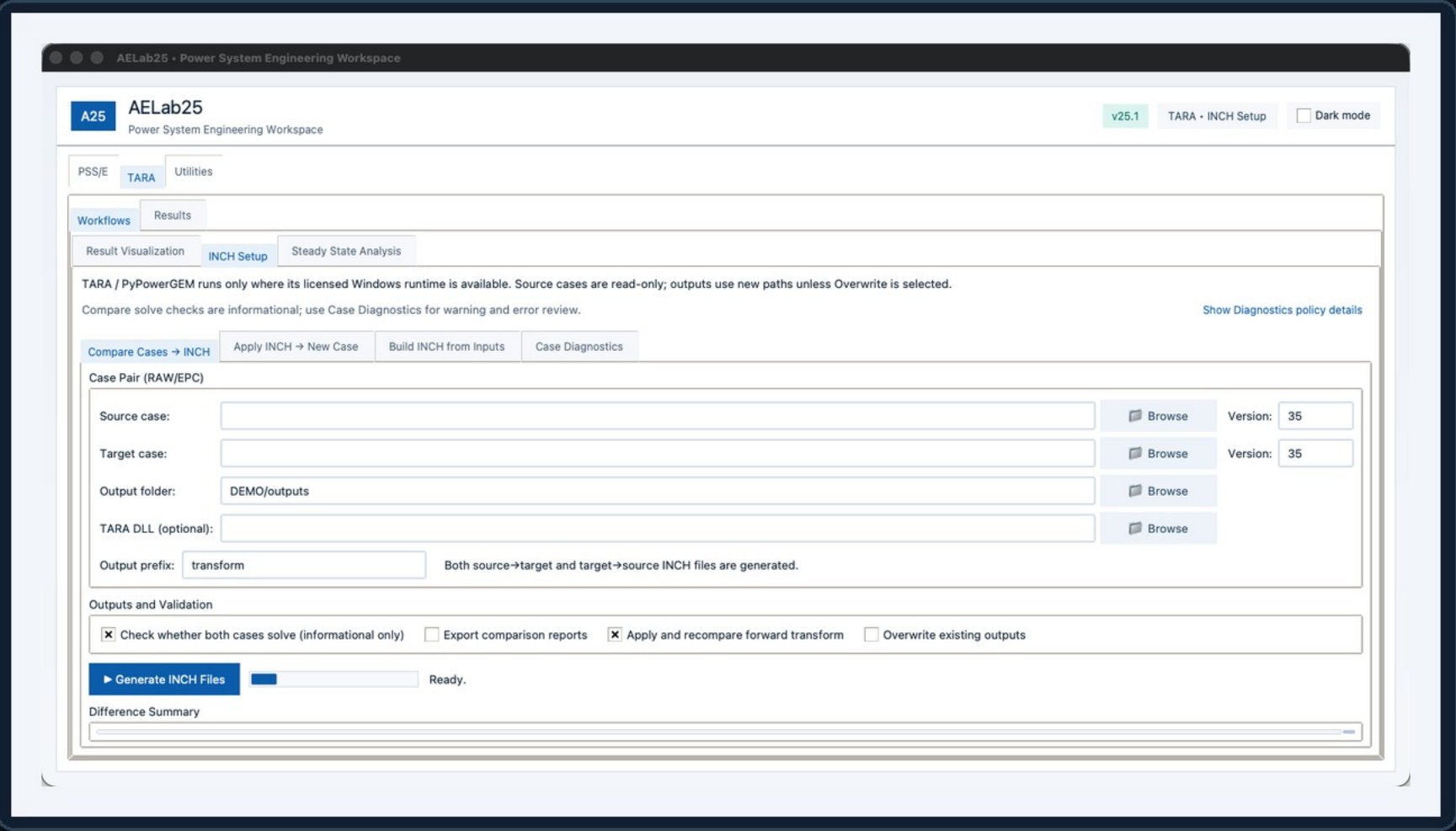
## WHAT THIS TAB DOES

Cross-compares transfer-limit batches and exposes filter and dashboard output settings.

REAL UI CAPTURE

# Compare Cases → INCH

TARA › Workflows › INCH Setup › Compare Cases → INCH



## WHAT THIS TAB DOES

Compares a source and target case and prepares forward and reverse INCH transformations with optional validation reports.

REAL UI CAPTURE

# Apply INCH → New Case

TARA › Workflows › INCH Setup › Apply INCH → New Case

AE Lab25 • Power System Engineering Workspace

A25

AE Lab25

Power System Engineering Workspace

v25.1TARA • INCH SetupDark mode

PSS/E

TARA

Utilities

Workflows

Results

Result Visualization

INCH Setup

Steady State Analysis

TARA / PyPowerGEM runs only where its licensed Windows runtime is available. Source cases are read-only; outputs use new paths unless Overwrite is selected.

Compare solve checks are informational; use Case Diagnostics for warning and error review.

[Show Diagnostics policy details](#)

Compare Cases → INCH

Apply INCH → New Case

Build INCH from Inputs

Case Diagnostics

Base and Output Cases (RAW/EPC)

Base case:

Browse

Version: 35

New case:

Browse

Version: 35

TARA DLL (optional):

Browse

Ordered INCH Stack (applied top to bottom)

Add INCH...

Remove

Move Up

Move Down

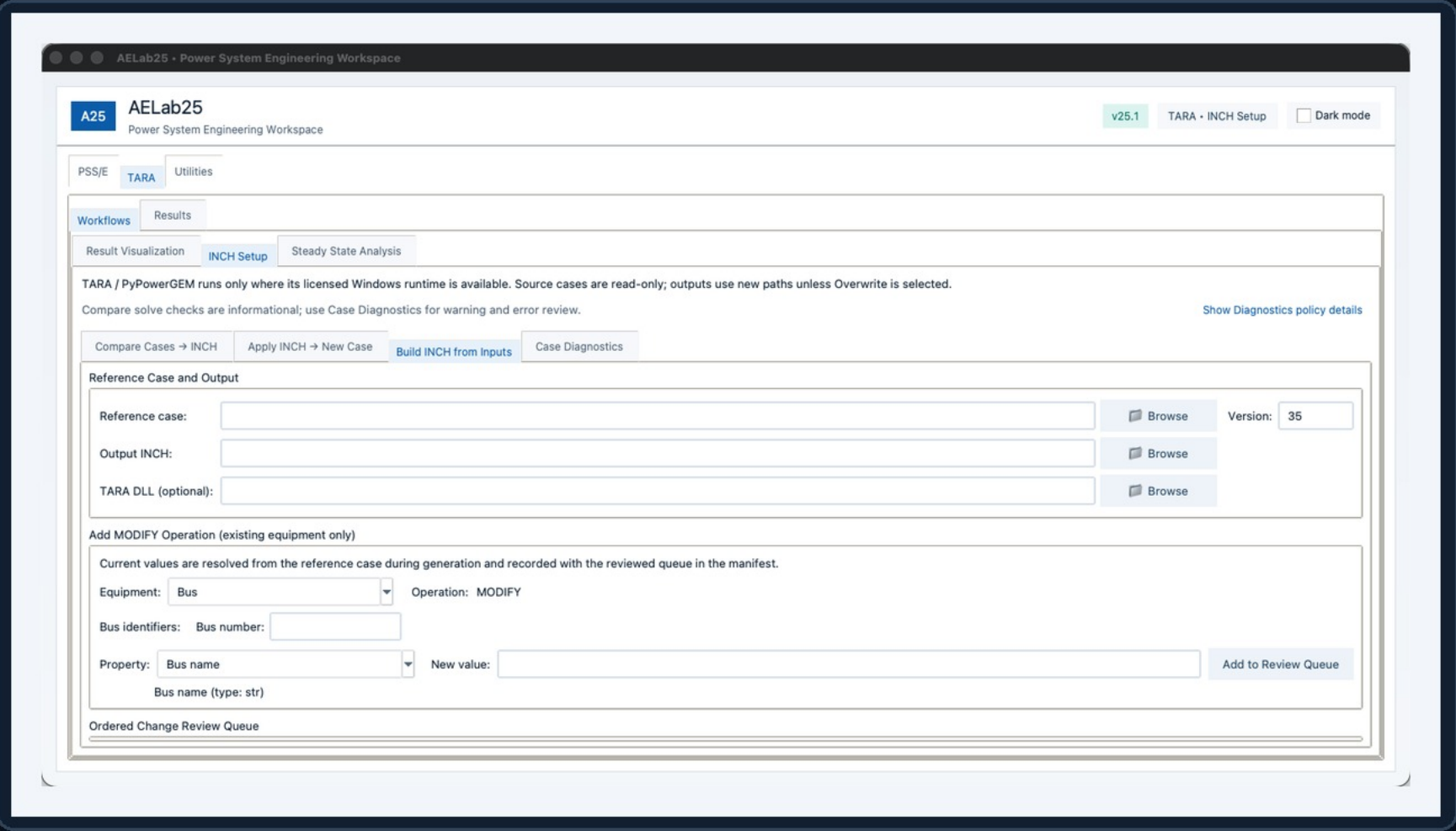
## WHAT THIS TAB DOES

Applies an ordered INCH stack to a base case and exposes solve, convergence, overwrite, and reopen checks.

## REAL UI CAPTURE

# Build INCH from Inputs

TARA › Workflows › INCH Setup › Build INCH from Inputs



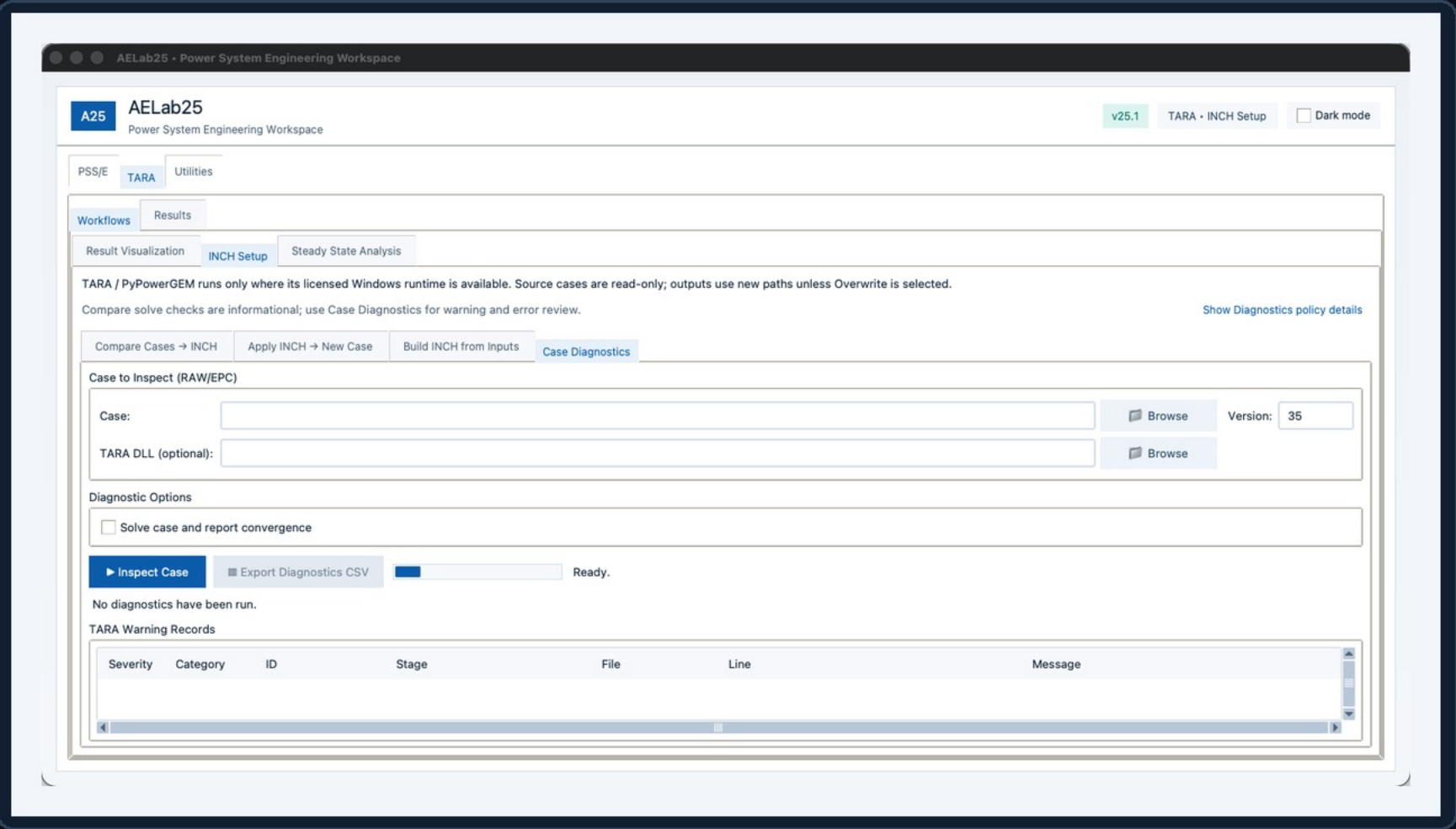
## WHAT THIS TAB DOES

Builds a reviewed, ordered queue of supported equipment modifications before writing an INCH file.

REAL UI CAPTURE

# Case Diagnostics

TARA › Workflows › INCH Setup › Case Diagnostics



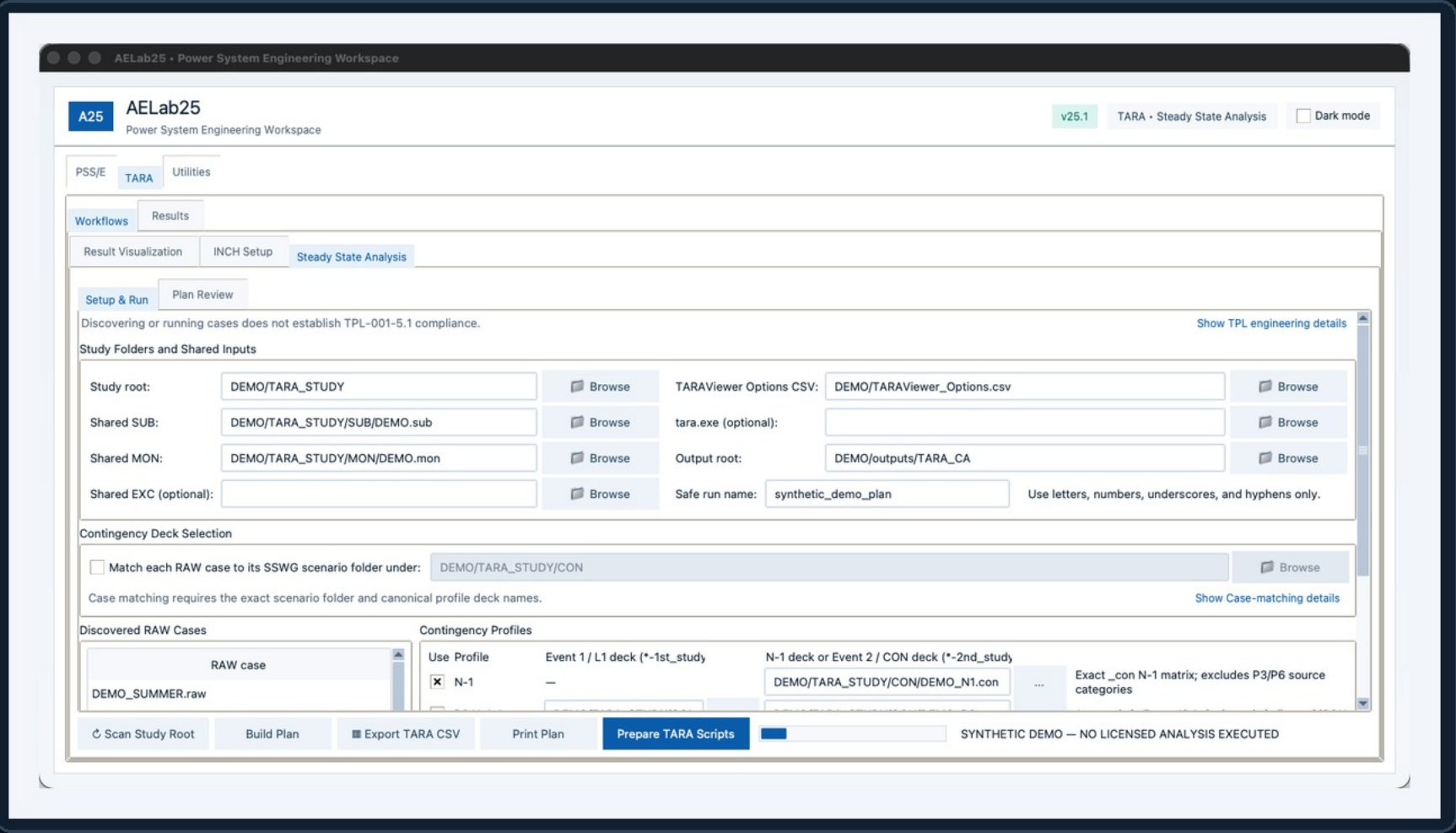
## WHAT THIS TAB DOES

Inspects a selected case and reports diagnostic warning records and optional convergence information.

REAL UI CAPTURE

# Steady State Analysis — Setup & Run

TARA › Workflows › Steady State Analysis › Setup & Run



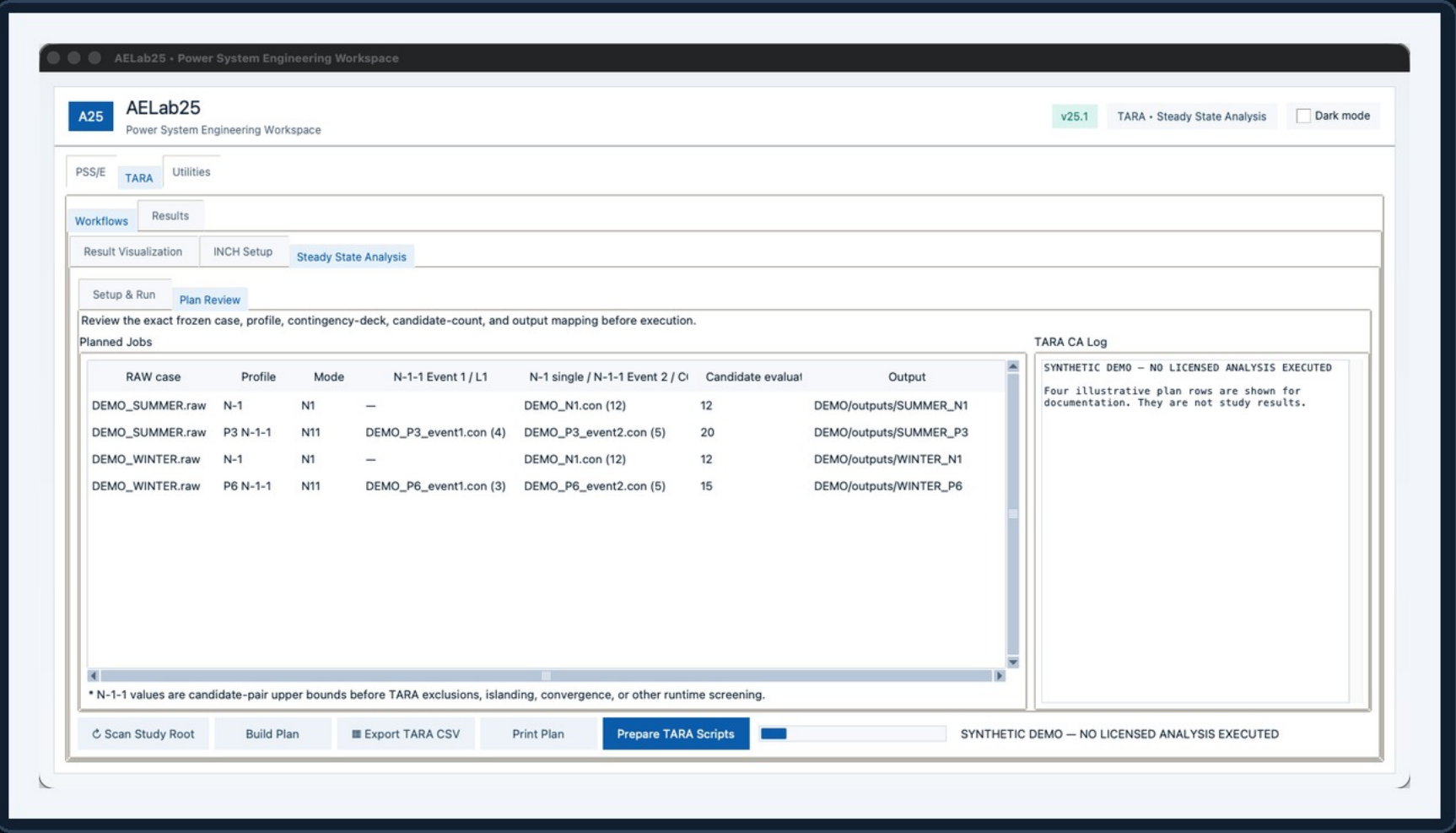
## WHAT THIS TAB DOES

Discovers RAW cases, maps contingency profiles, configures analysis options, and prepares or executes the resulting job plan.

REAL UI CAPTURE

# Steady State Analysis — Plan Review

TARA › Workflows › Steady State Analysis › Plan Review



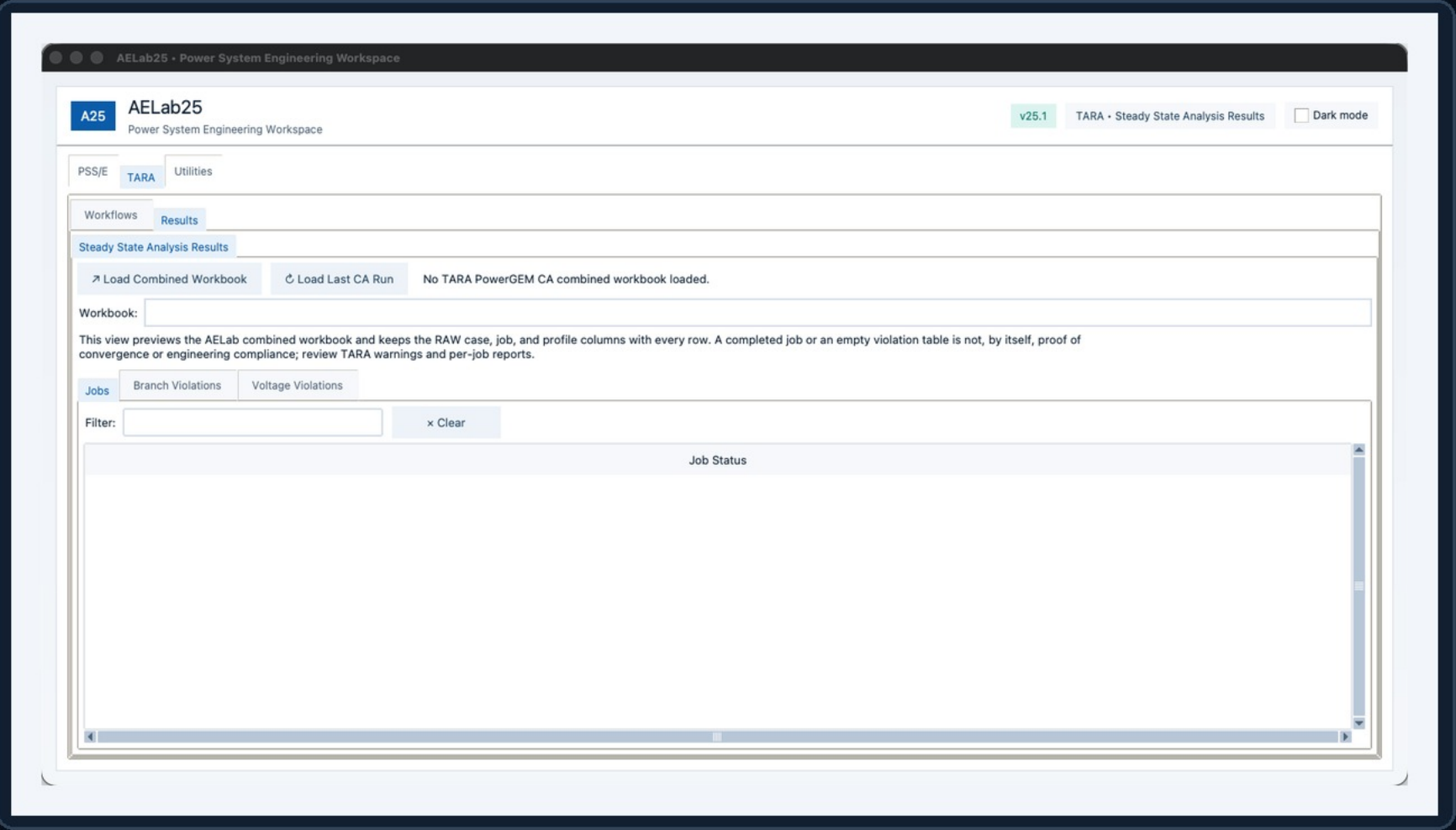
## WHAT THIS TAB DOES

Reviews the frozen job plan, per-job commands and dependencies, and the CSV export and printable plan controls.

REAL UI CAPTURE

# TARA Jobs

TARA › Results › Steady State Analysis Results › Jobs



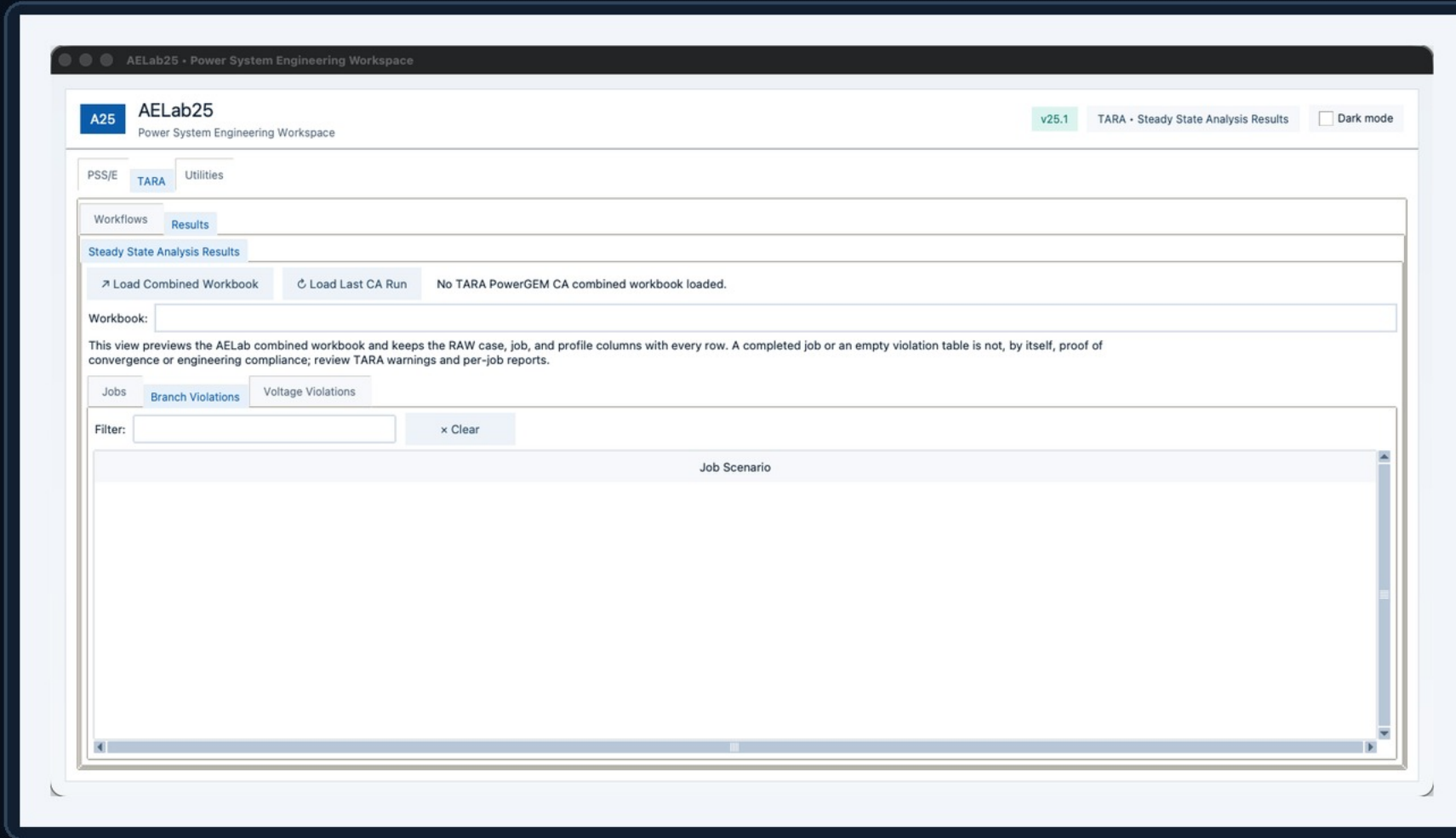
## WHAT THIS TAB DOES

Reviews combined-workbook job status while retaining source case and contingency-profile context.

REAL UI CAPTURE

# TARA Branch Violations

TARA › Results › Steady State Analysis Results › Branch Violations



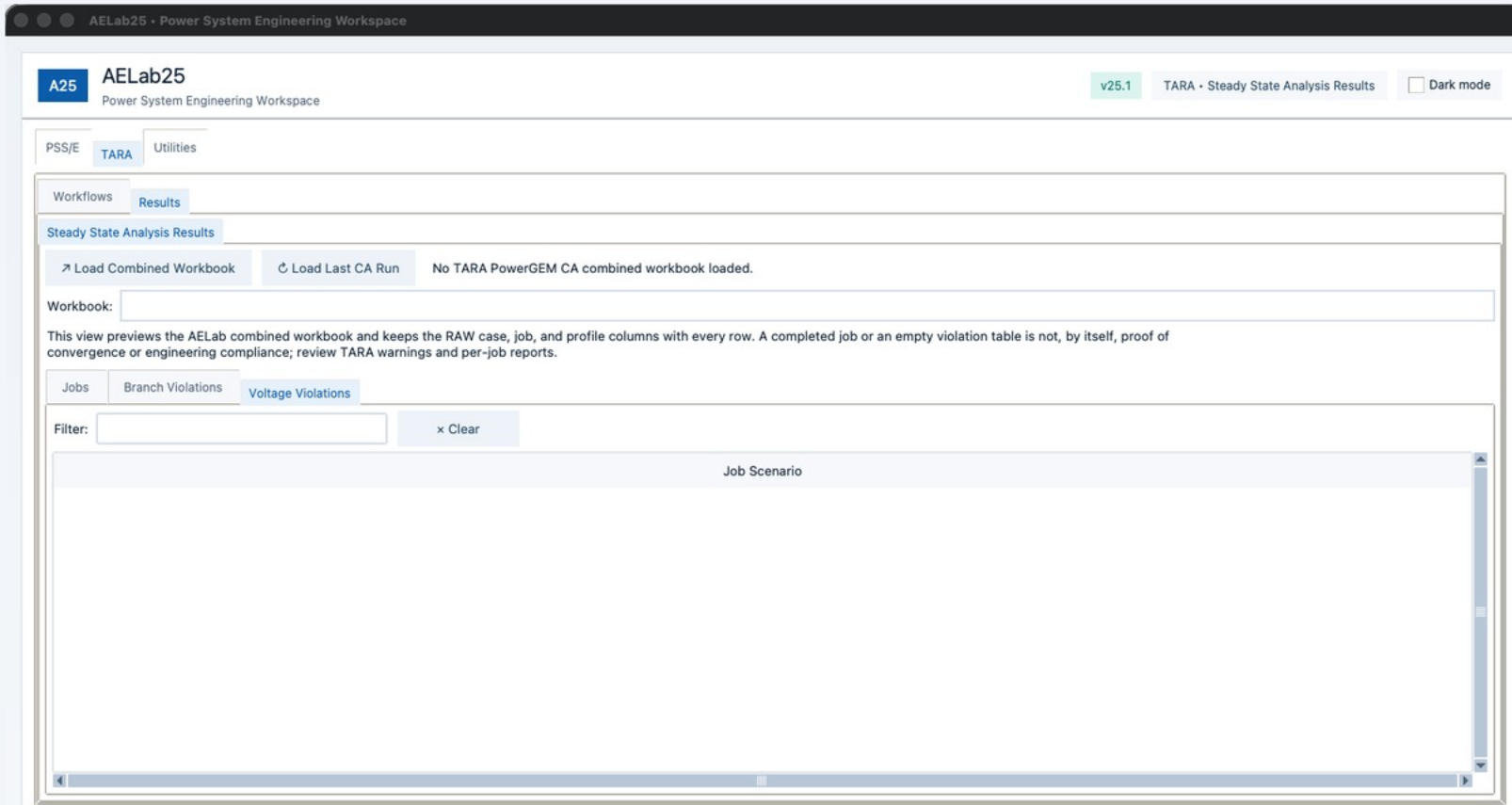
## WHAT THIS TAB DOES

Reviews reported branch violations together with their originating job and scenario fields.

REAL UI CAPTURE

# TARA Voltage Violations

TARA › Results › Steady State Analysis Results › Voltage Violations



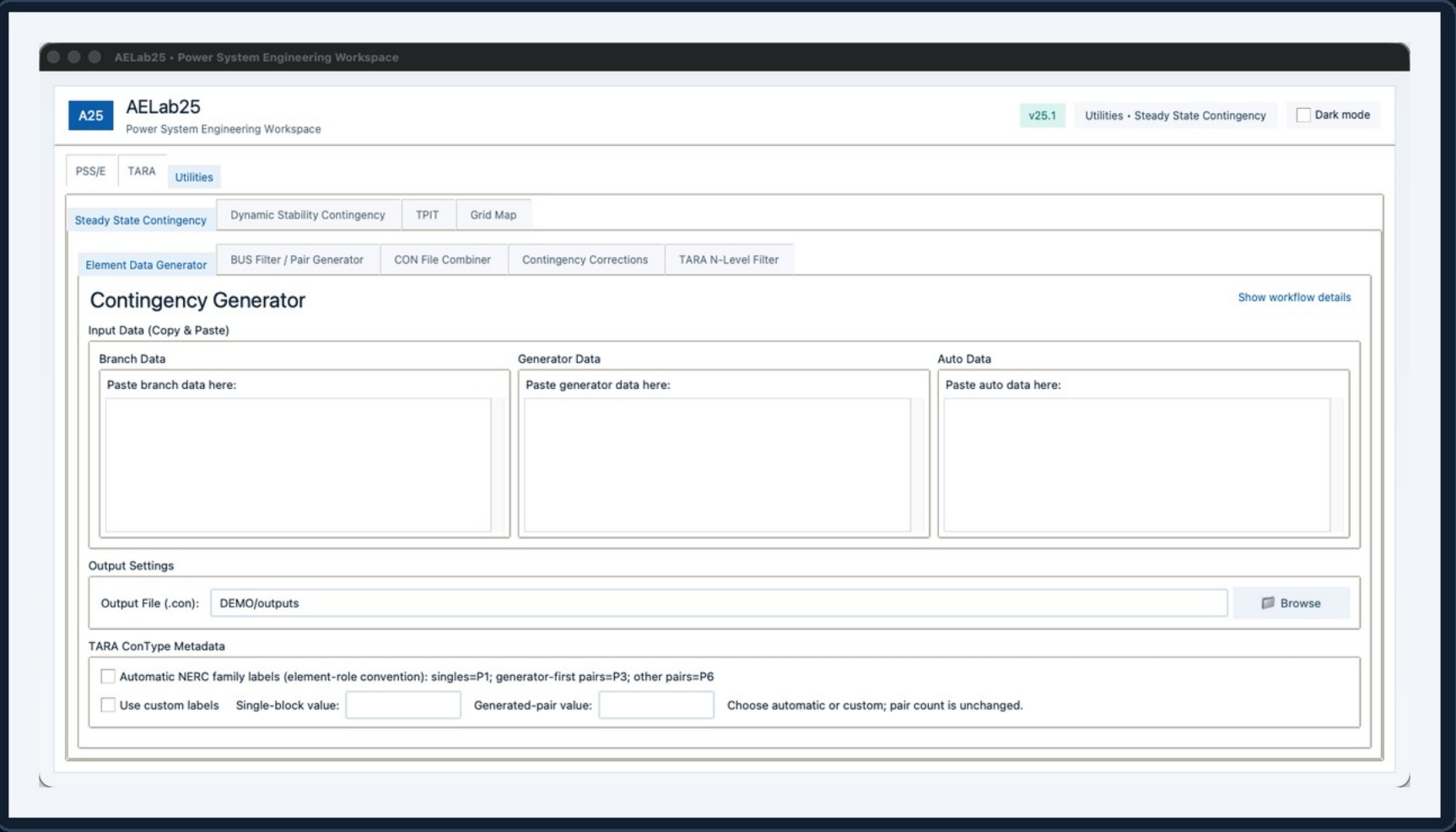
## WHAT THIS TAB DOES

Reviews reported voltage violations together with their originating job and scenario fields.

REAL UI CAPTURE

# Element Data Generator

Utilities › Steady State Contingency › Element Data Generator



## WHAT THIS TAB DOES

Creates single and paired contingency blocks from pasted branch, generator, and transformer data with optional ConType metadata.

REAL UI CAPTURE

# BUS Filter / Pair Generator

Utilities › Steady State Contingency › BUS Filter / Pair Generator

The screenshot shows the 'BUS Filter / Pair Generator' tab within the 'Utilities' section of the 'Steady State Contingency' workflow. The interface includes a header with 'AELab25 Power System Engineering Workspace', version 'v25.1', and a 'Dark mode' toggle. The main content area has a sidebar with 'PSS/E', 'TARA', and 'Utilities' tabs. Under 'Utilities', there are sub-tabs for 'Steady State Contingency', 'Dynamic Stability Contingency', 'TPIT', and 'Grid Map'. The 'Steady State Contingency' sub-tab is active, showing further sub-tabs: 'Element Data Generator', 'BUS Filter / Pair Generator' (selected), 'CON File Combiner', 'Contingency Corrections', and 'TARA N-Level Filter'. The 'BUS Filter / Pair Generator' sub-tab contains the following sections:

- Input and Output Files:** Two input fields with 'Browse' buttons. The first is 'Source Contingency File (.con):' and the second is 'Output File (.con):' with the value 'DEMO/outputs'.
- Optional Custom TARA Report Labels:** A checkbox 'Write set types ConType / set values metadata' is unchecked. To its right are two input fields: 'Source-block value:' and 'Generated-pair value:', followed by the text 'Manual labels only.'
- Bus Numbers:** A text area with the instruction 'Paste bus numbers here. Newline, comma, semicolon, tab, and space separated lists are supported.'

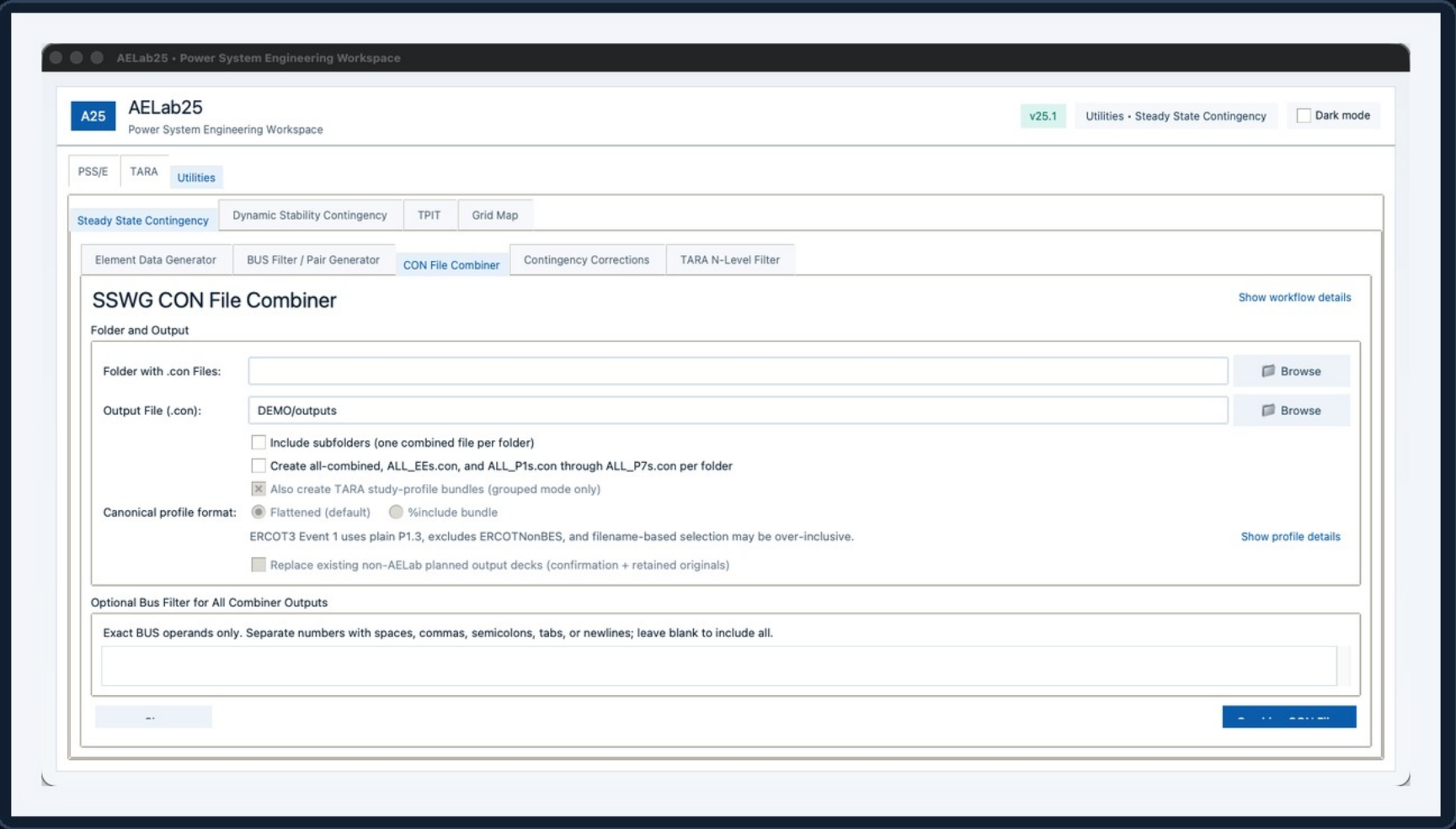
## WHAT THIS TAB DOES

Filters source contingency blocks by a reviewed bus list and can create matched pair combinations.

REAL UI CAPTURE

# CON File Combiner

Utilities › Steady State Contingency › CON File Combiner



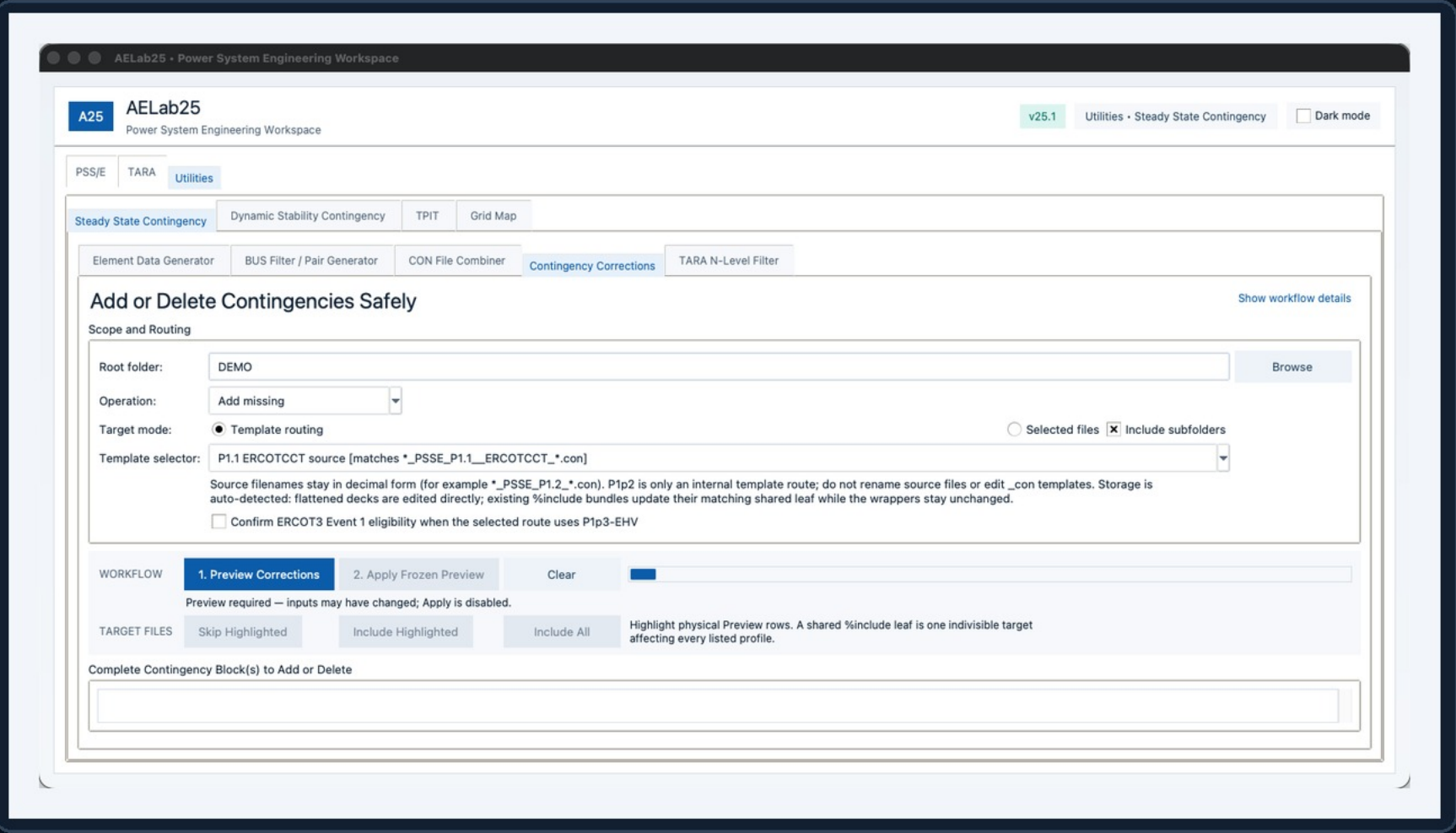
## WHAT THIS TAB DOES

Combines contingency decks and can prepare grouped NERC-family and TARA study-profile outputs.

REAL UI CAPTURE

# Contingency Corrections

Utilities › Steady State Contingency › Contingency Corrections



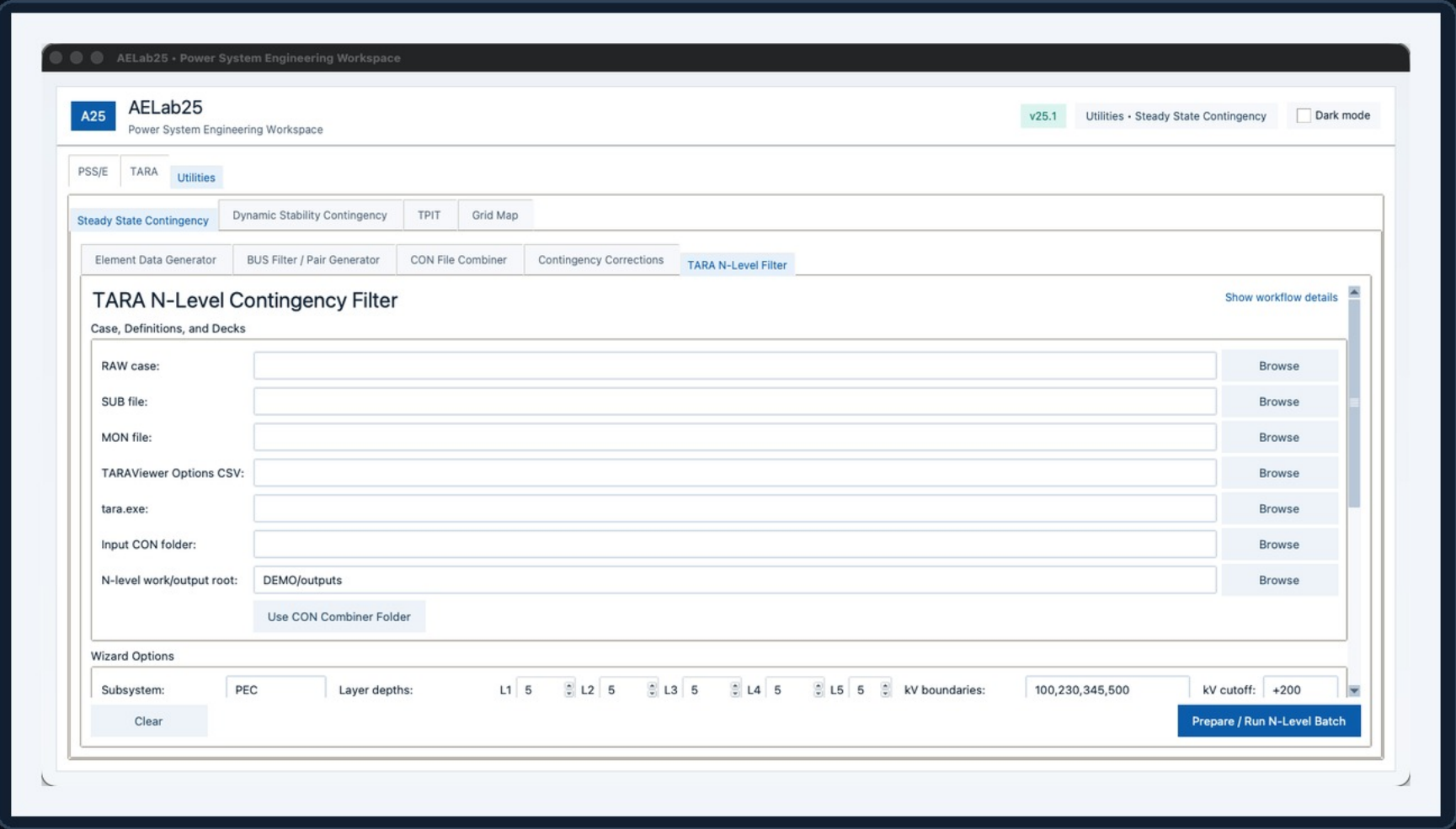
## WHAT THIS TAB DOES

Scans selected decks and requires a frozen preview before adding or deleting complete contingency blocks.

REAL UI CAPTURE

# TARA N-Level Filter

Utilities › Steady State Contingency › TARA N-Level Filter



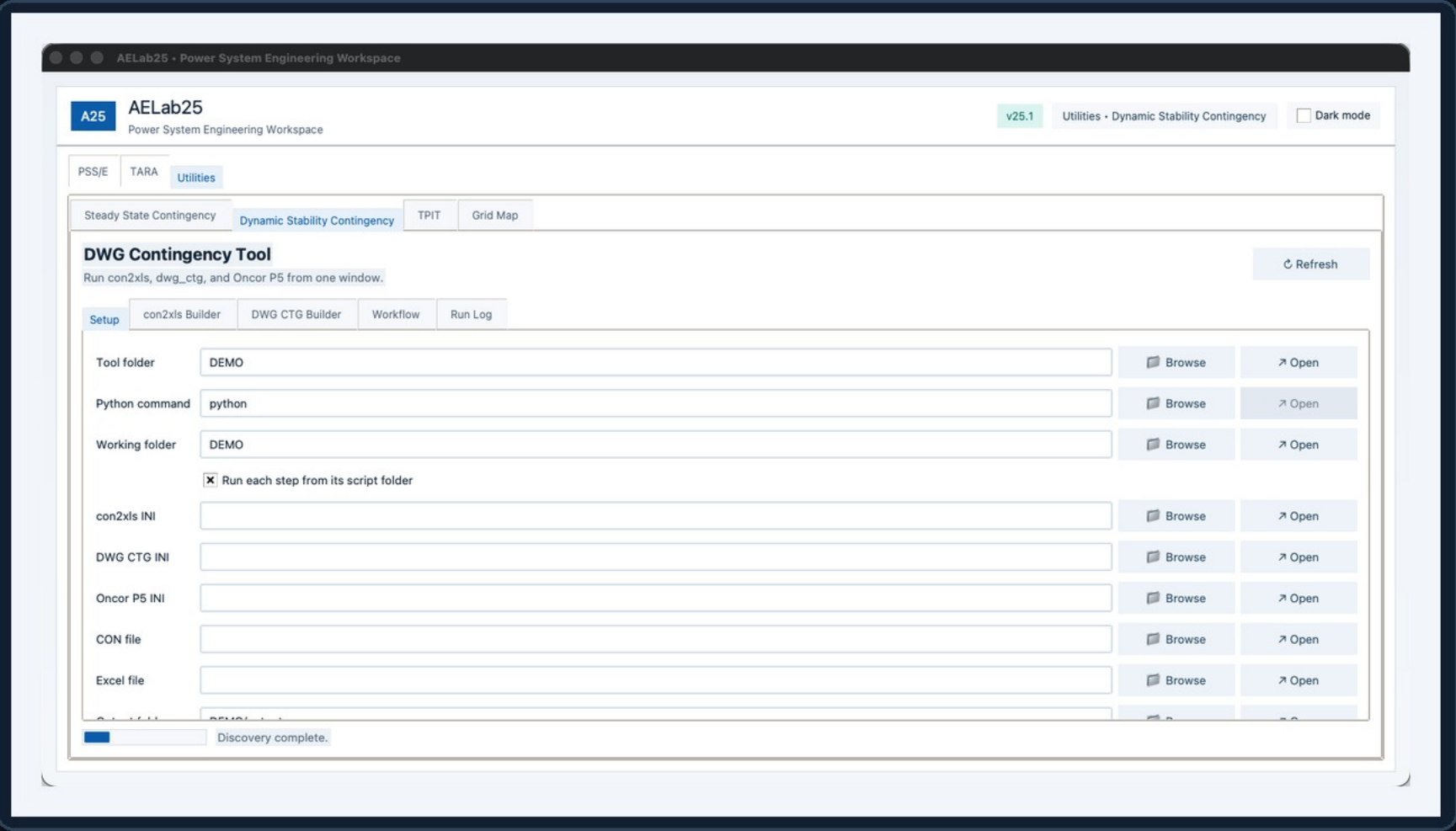
## WHAT THIS TAB DOES

Configures subsystem, layer, kV, generator, and deck-scope filters and can prepare scripts without launching licensed TARA.

REAL UI CAPTURE

# DWG Contingency — Setup

Utilities › Dynamic Stability Contingency › Setup



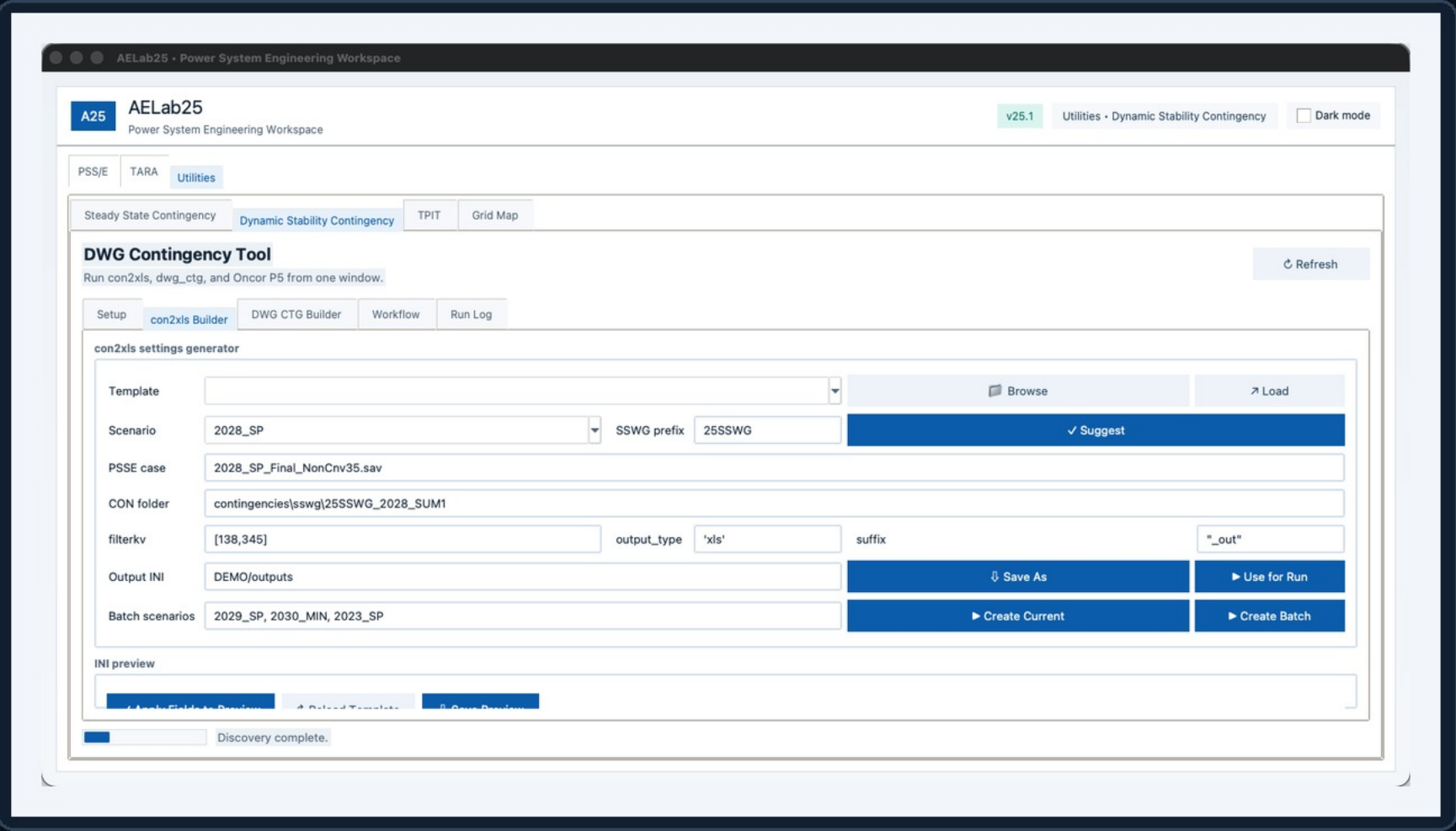
## WHAT THIS TAB DOES

Locates the DWG toolchain, configures working inputs and outputs, and reports discovery status.

REAL UI CAPTURE

# con2xls Builder

Utilities › Dynamic Stability Contingency › con2xls Builder



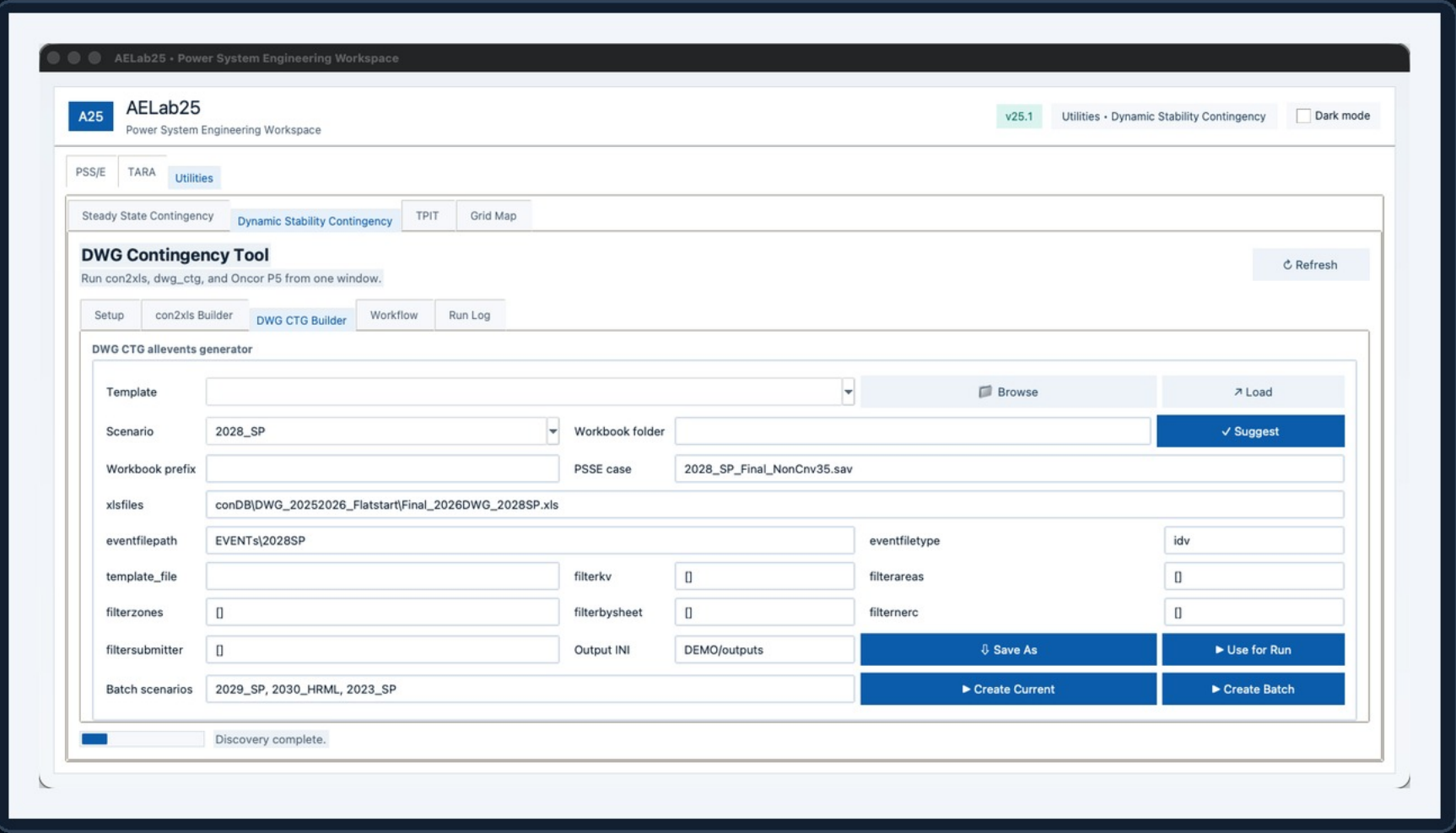
## WHAT THIS TAB DOES

Builds reviewed con2xls settings from a template and scenario-specific workbook inputs.

REAL UI CAPTURE

# DWG CTG Builder

Utilities › Dynamic Stability Contingency › DWG CTG Builder



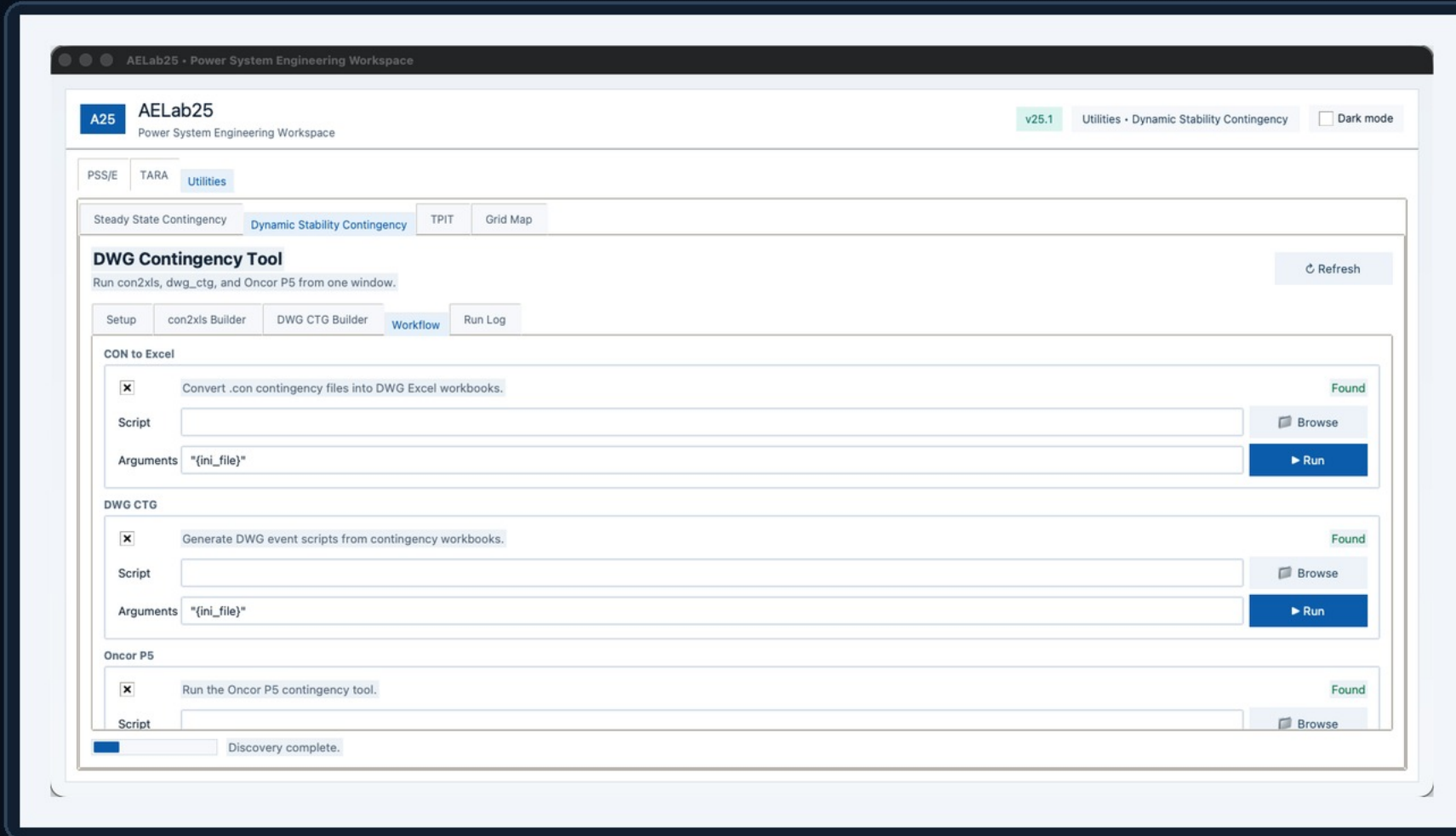
## WHAT THIS TAB DOES

Builds DWG contingency settings for selected workbooks, event files, and optional filters.

REAL UI CAPTURE

# DWG Contingency — Workflow

Utilities › Dynamic Stability Contingency › Workflow



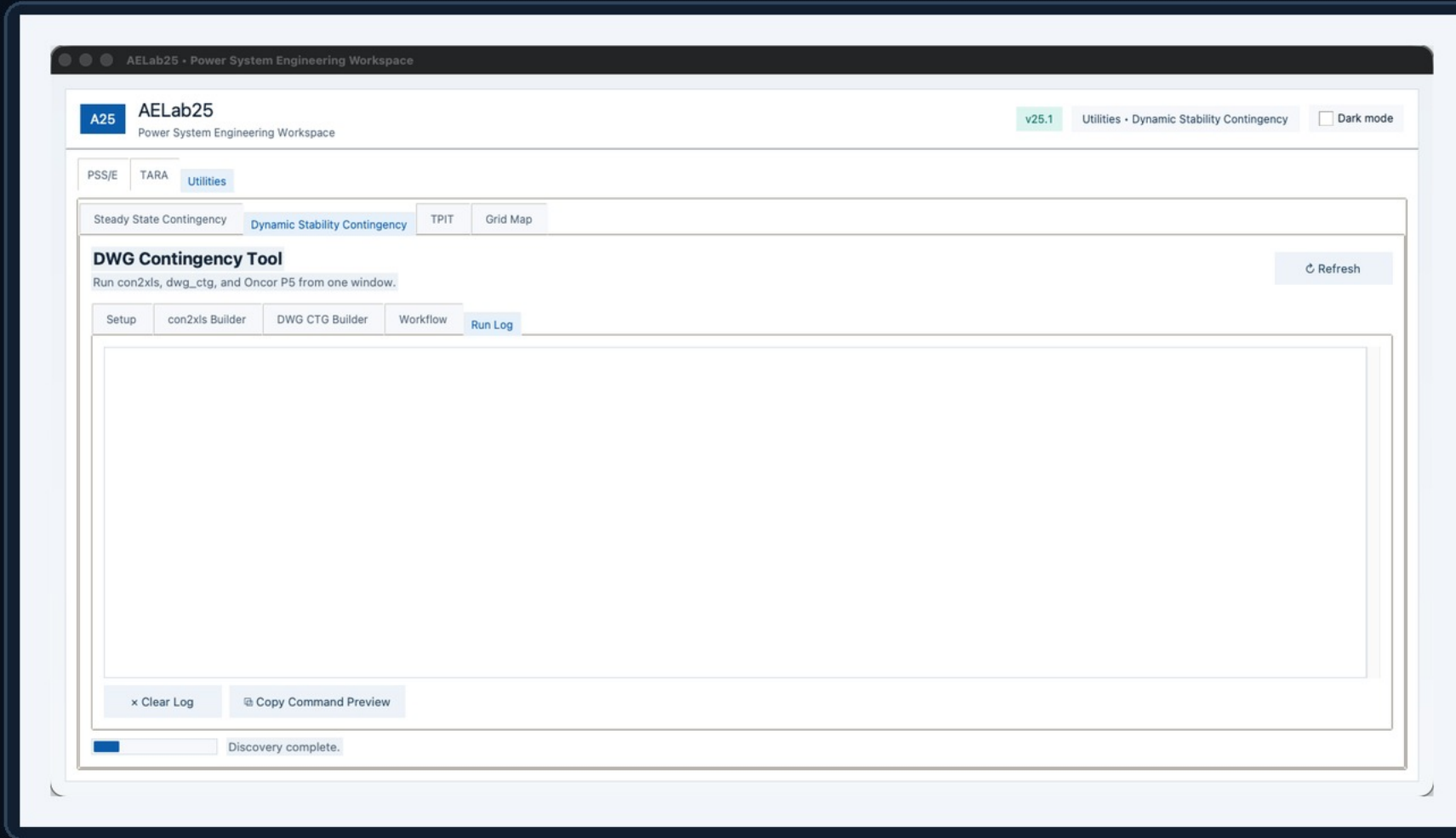
## WHAT THIS TAB DOES

Shows ordered con2xls, DWG contingency, and Oncor P5 workflow steps with individual run selection and status.

REAL UI CAPTURE

# DWG Contingency — Run Log

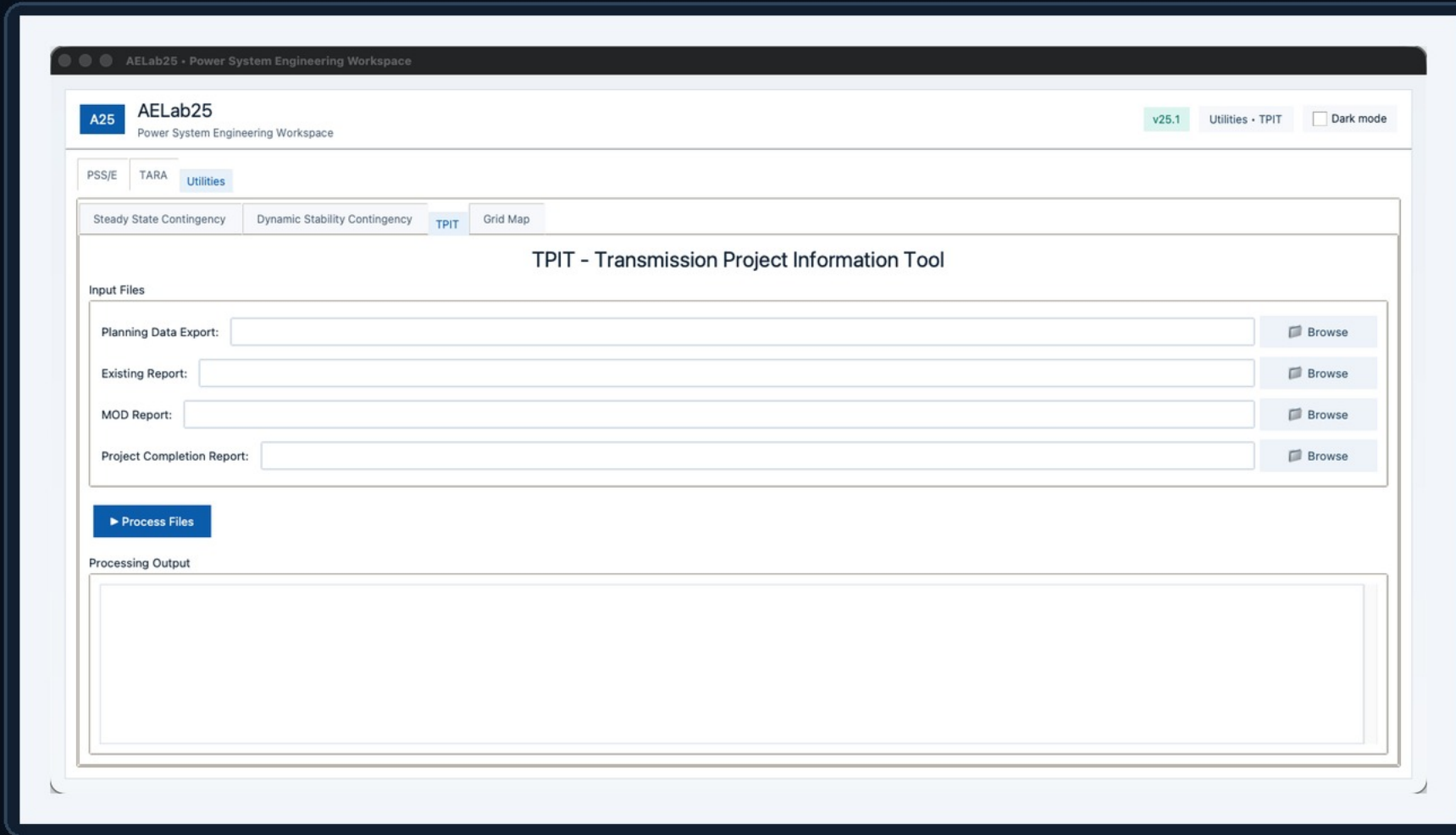
Utilities › Dynamic Stability Contingency › Run Log



## WHAT THIS TAB DOES

Displays workflow messages and subprocess output while keeping failures visible.

REAL UI CAPTURE



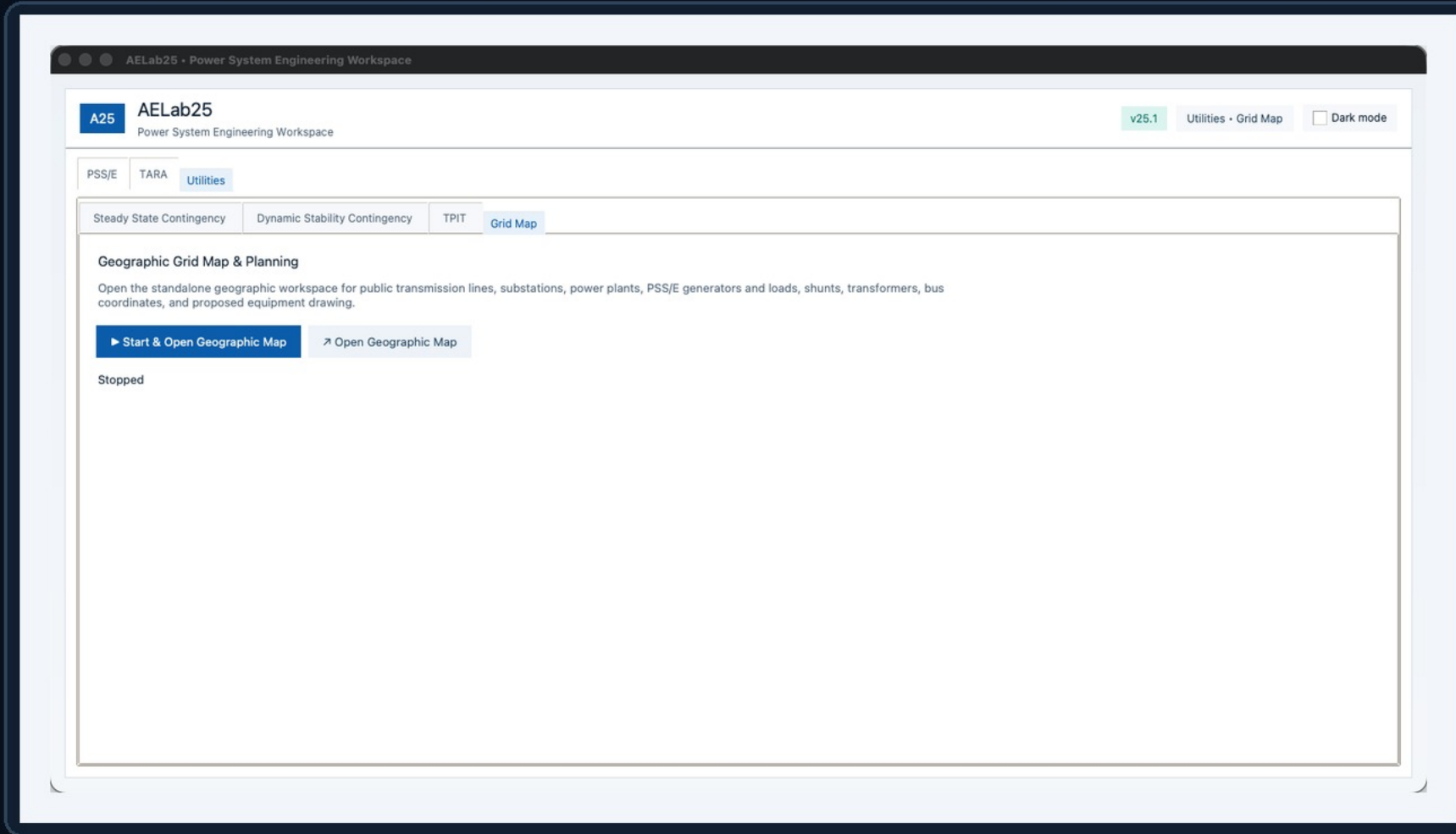
## WHAT THIS TAB DOES

Loads planning, existing, MOD, and completion reports for transmission-project information processing.

REAL UI CAPTURE

# Grid Map

Utilities › Grid Map



## WHAT THIS TAB DOES

Configures the geographic planning map and its local interactive server controls.

REAL UI CAPTURE